ROSMASTER-M1PRO User Manual

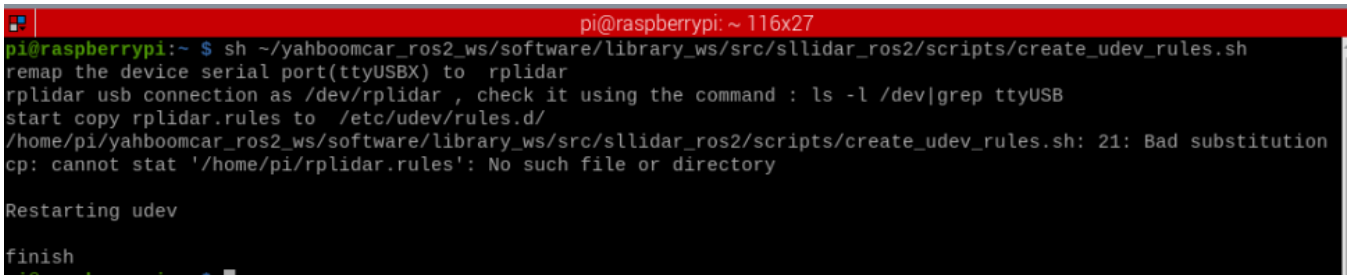
ROSMASTER-M1PRO User Manual

- 0. Preparations
- 1. First Use
  - 1.1 Enter the Configuration Wizard
  - 1.2 Select the Hardware Configuration
  - 1.3 Quick Start of Gamepad Control
- 2. How to Connect to the Vehicle
  - 2.1 VNC Connection
  - 2.2 SSH Connection
- 3. Connecting the Vehicle to Wi-Fi
  - 3.1 How to Obtain the Robot Vehicle IP Address
  - 3.2 Connecting the Robot Vehicle to an External Wi-Fi
    - Method 1: Connect via the VNC graphical interface with an Ethernet cable (Recommended)
    - Method 2: Without an Ethernet cable, configure Wi-Fi via the hotspot IP and modify the Wi-Fi priority for automatic connection
- 4. Tutorial Symbol Description
- 5. Viewing Source Code and Building Packages
  - 5.1 Viewing Code
  - 5.2 Building Packages

0. Preparations

- Refer to 【Robot Configuration and Operation Guide — Assembly Videos】 to complete the assembly of the vehicle components and power on the vehicle.
- If the selected configuration is **Slamtec C1 LiDAR**, you need to first refer to **2.1 VNC Connection** to connect to the vehicle, bind the LiDAR rule file, and then proceed with the subsequent steps.

```
sh ~/yahboomcar_ros2_ws/software/library_ws/src/sllidar_ros2/scripts/create_udev_rules.sh
```



- After the prompt above appears, unplug and replug the Slamtec C1 LiDAR to complete the device port rule binding.

1. First Use

1.1 Enter the Configuration Wizard

- After the vehicle boots up, the OLED screen will display the vehicle hotspot `IP:192.168.1.11`



- Connect your phone or computer to the car's hotspot, open a browser, and access the initialization configuration wizard:

  ```
  http://192.168.1.11:8080
  ```

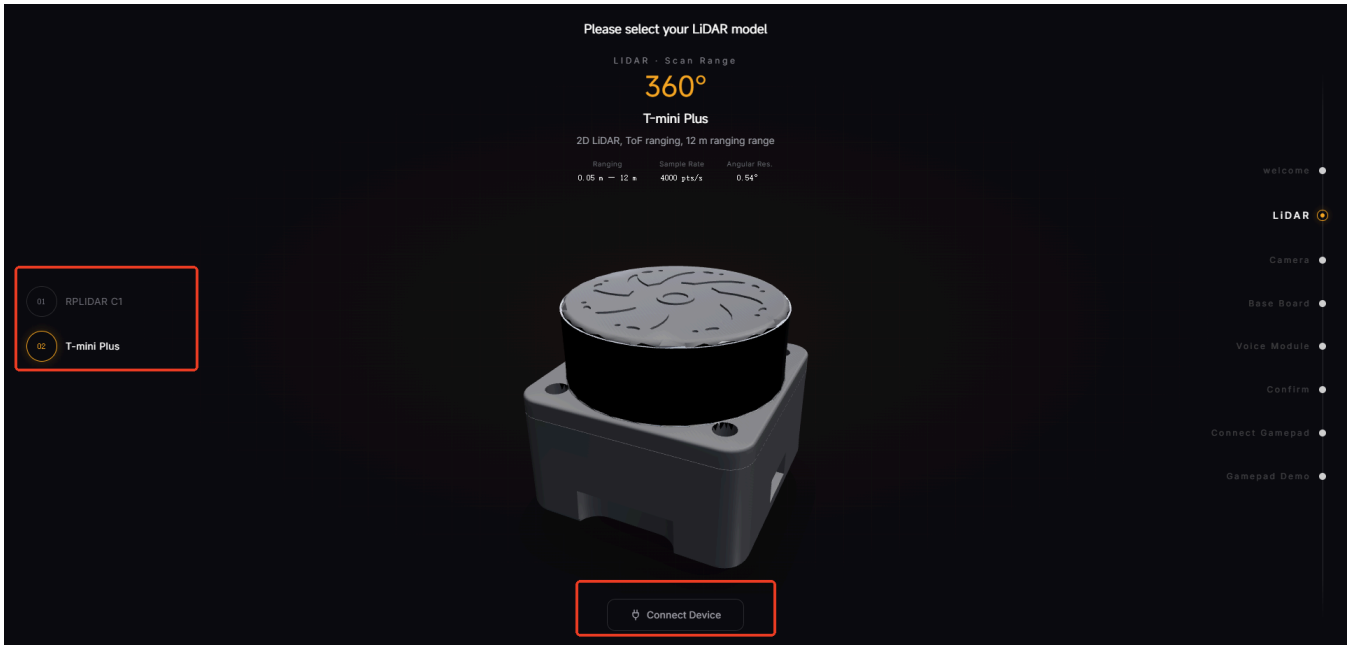
- You can switch the display language (Chinese/English) at the bottom of the interface.



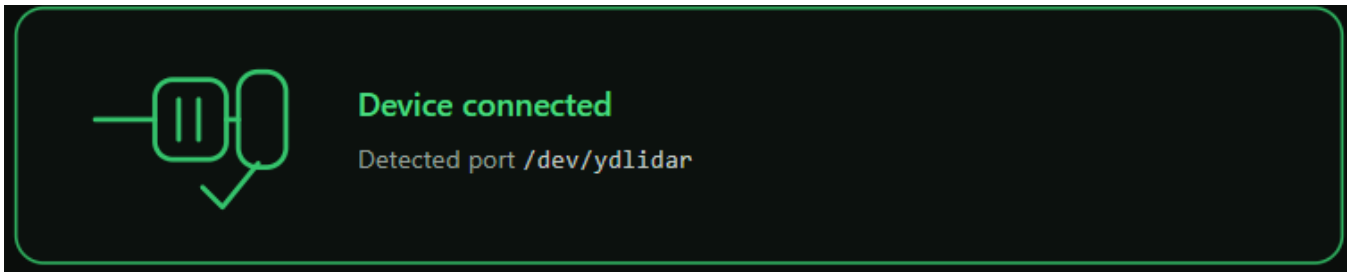
1.2 Select the Hardware Configuration

- ROSMASTER-M1PRO provides different cameras 【depth camera, USB monocular camera】and LiDARs 【Slamtec C1, Tmini Plus】 for selection. Select your camera and LiDAR version in the configuration wizard.

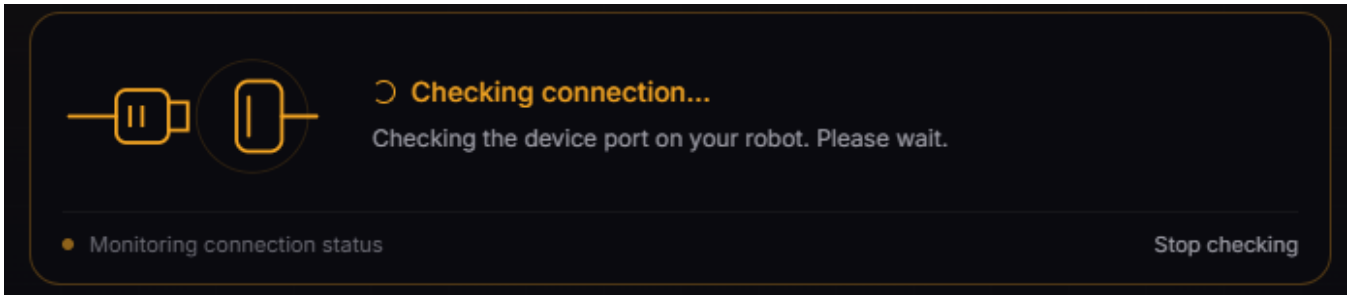
- For LiDAR selection, select your LiDAR model on the left, and then click Connect Device.



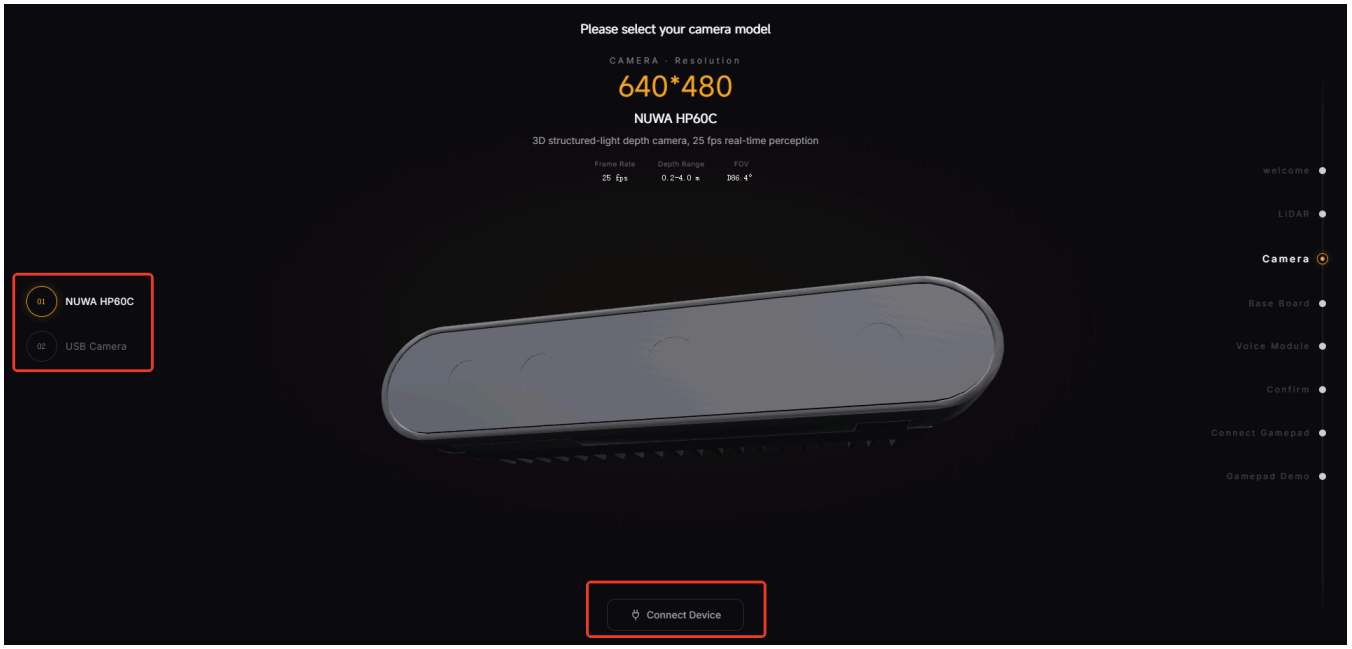
- If the device is wired correctly, a message indicating successful device connection will be displayed.



- If a message indicating waiting for device access is displayed, check whether the device wiring is loose, or unplug and replug the device.

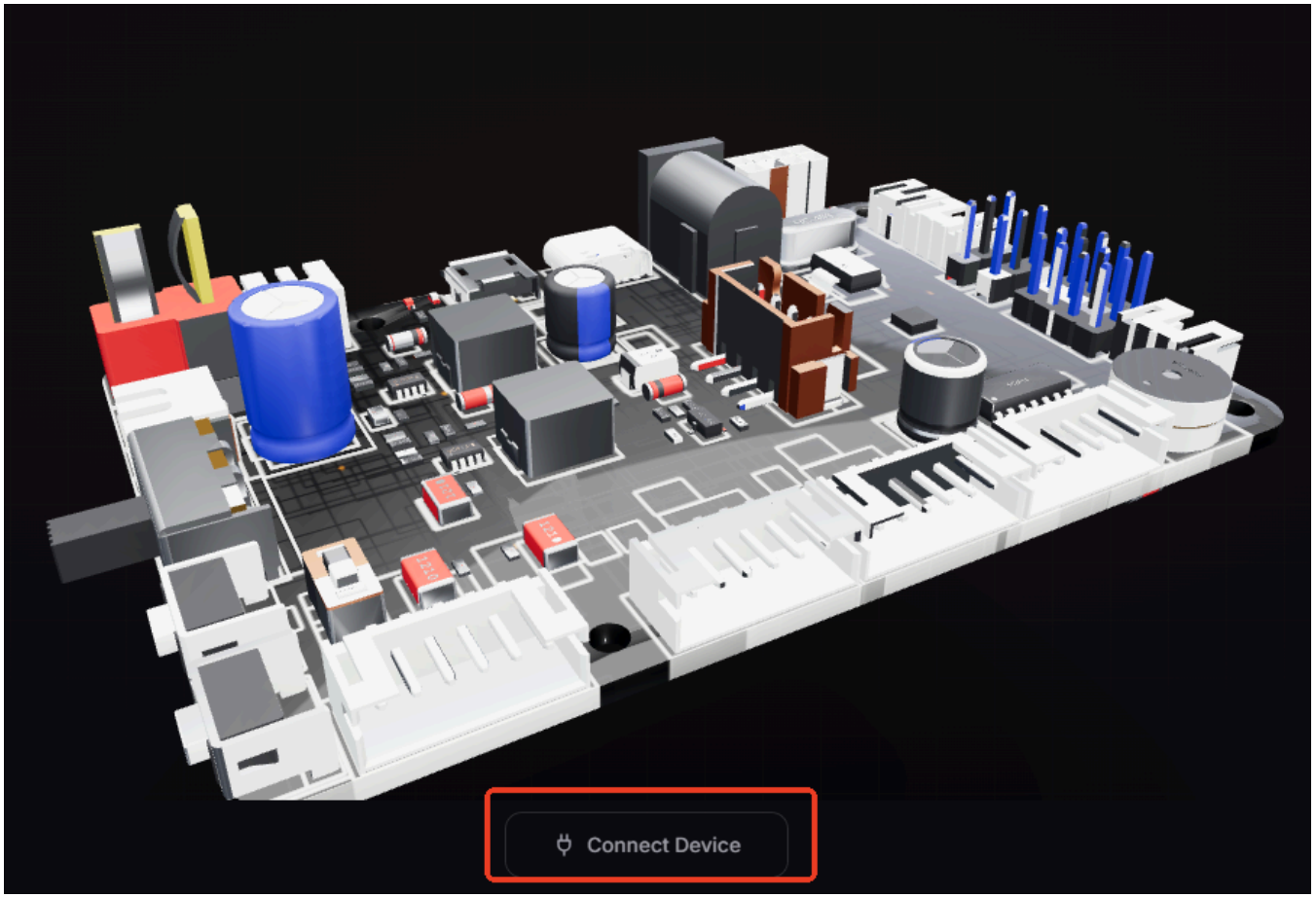


- For camera selection, perform the same operation: select the device on the left, and then click Connect Device.

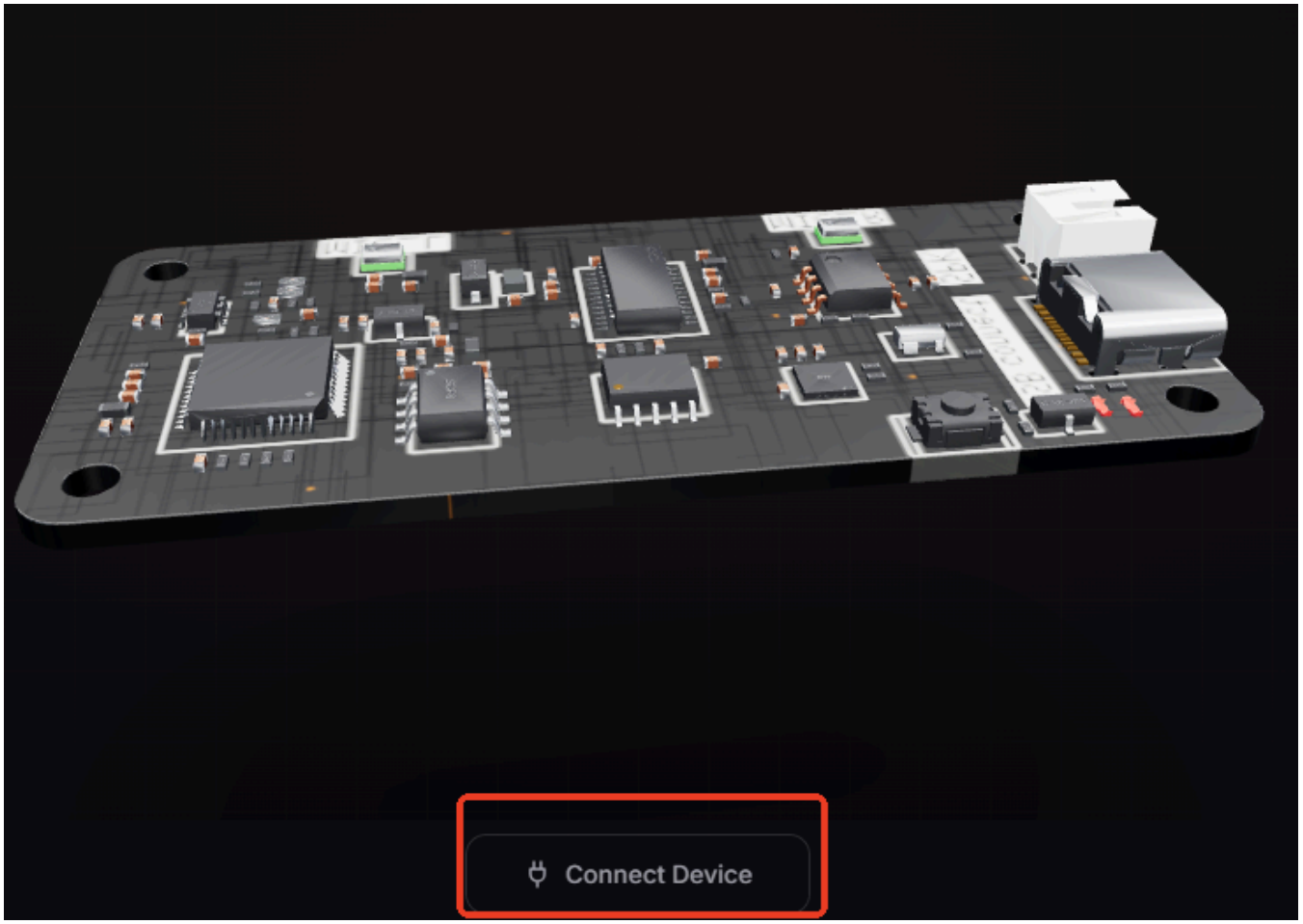


> **Note**
>
> The same applies to the other device pages. The remaining devices do not require model selection; simply connect the device to determine whether the device wiring is properly connected during assembly.

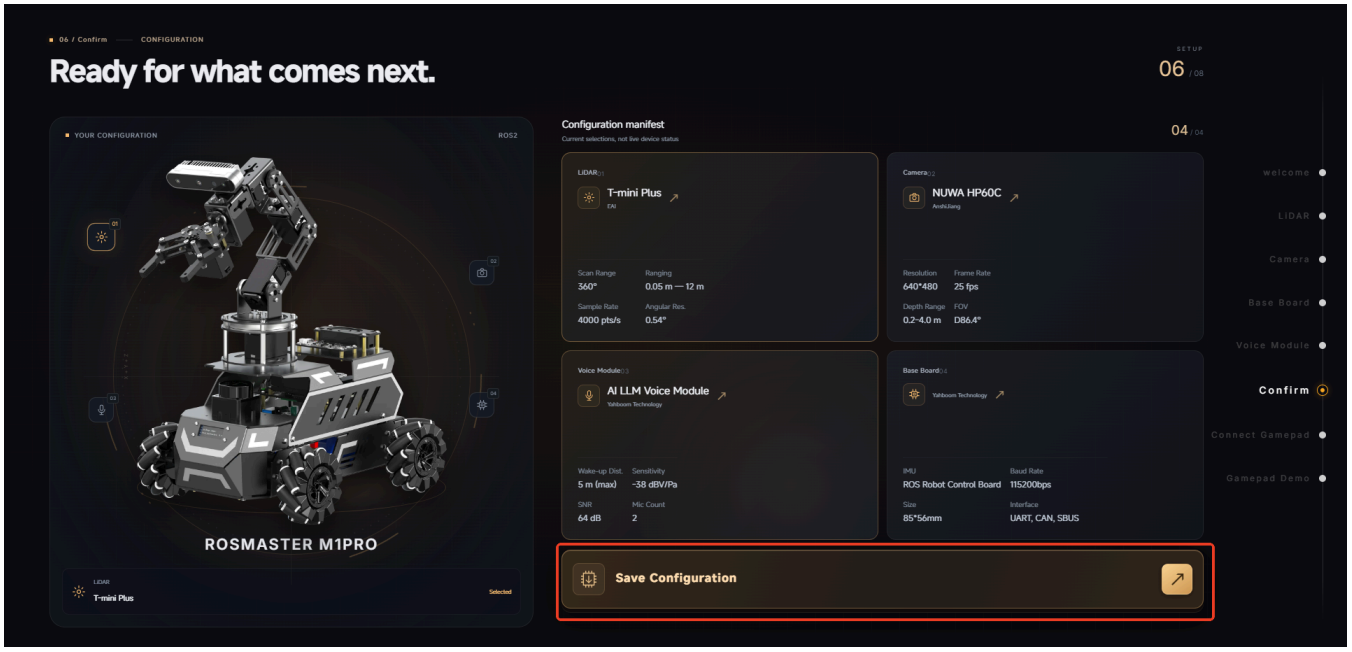
- For the robot control board, click Connect Device.



- For the voice module, click Connect Device.



- After completing the camera and LiDAR selection and confirming that the device wiring is correct, click Save Configuration.

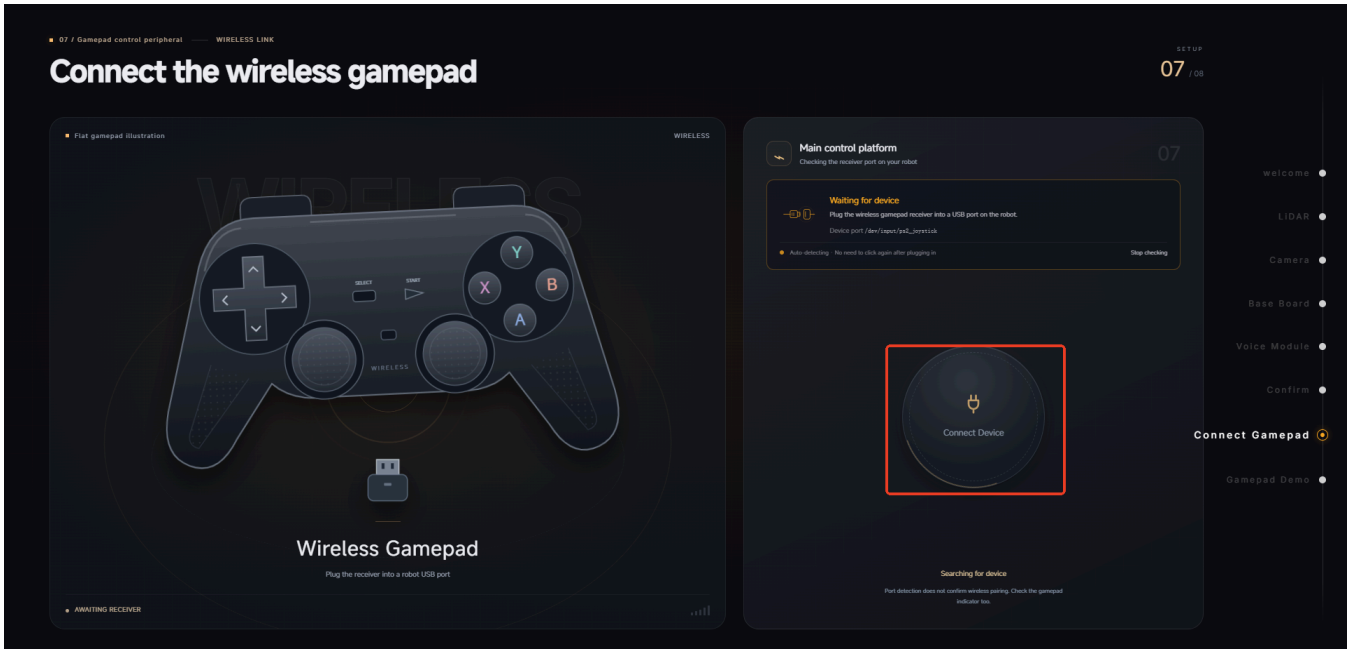




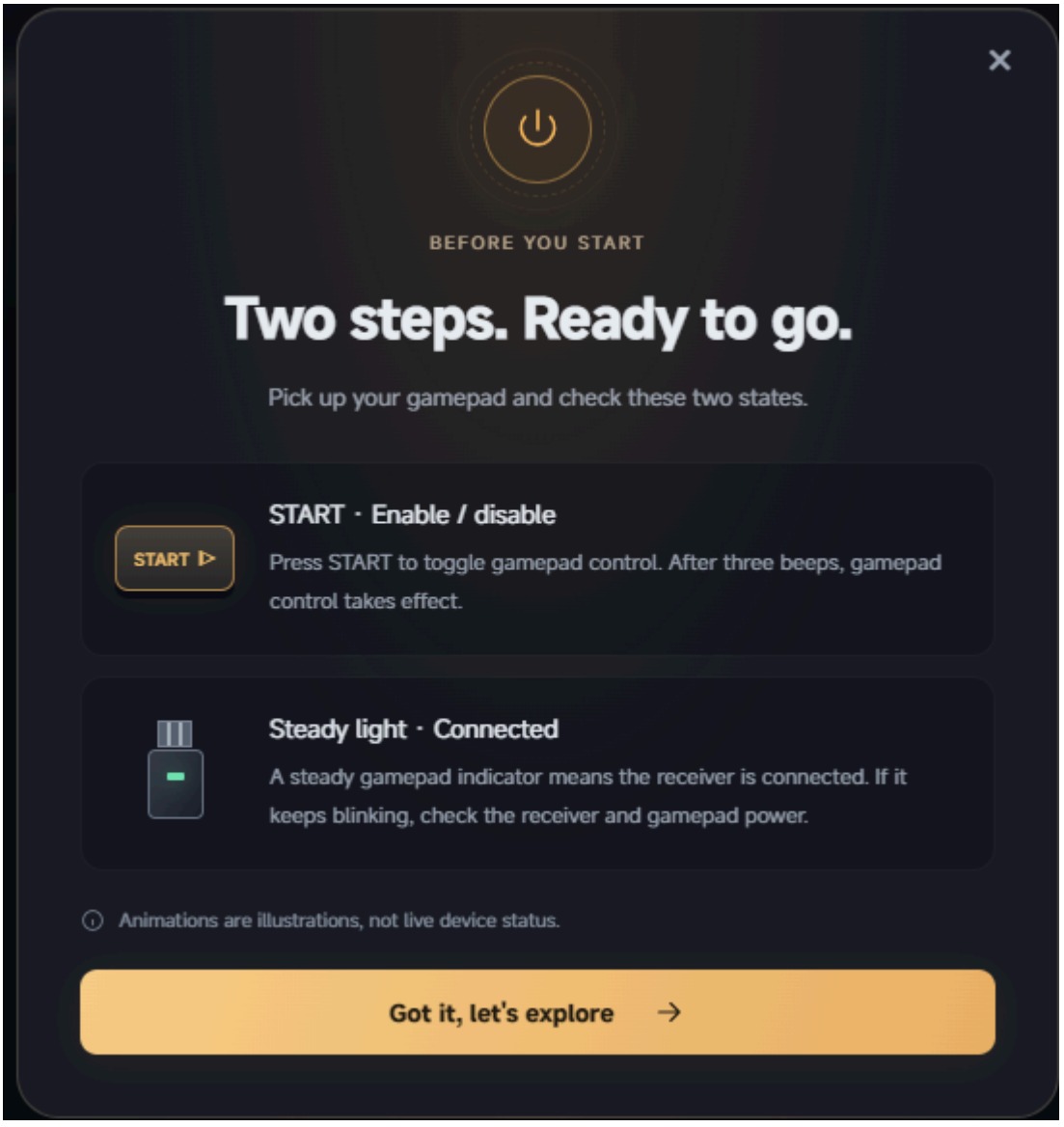
1.3 Quick Start of Gamepad Control

> **Tip**
>
> For first-time use, after completing the hardware configuration, insert the gamepad receiver into the USB port of the upper main controller, and then click Start Gamepad Control.

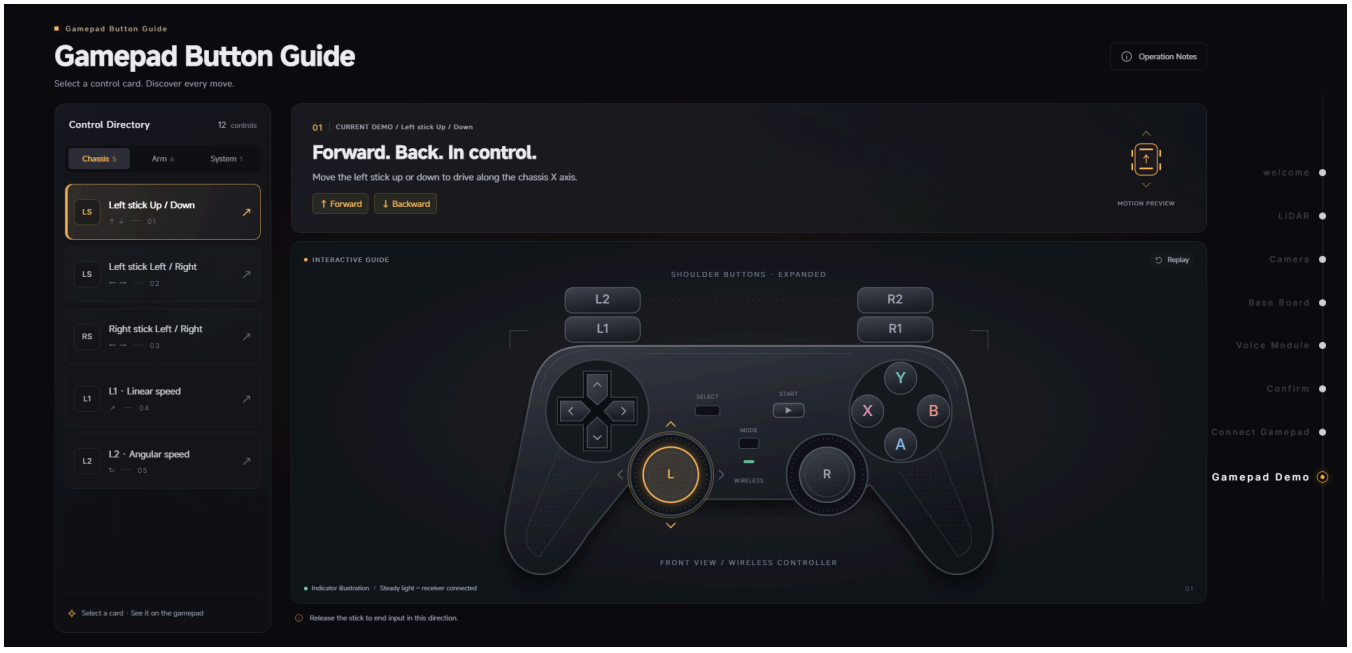
- Insert the gamepad receiver into the main control board. After the gamepad receiver shows that it is ready, click Start Gamepad Control.



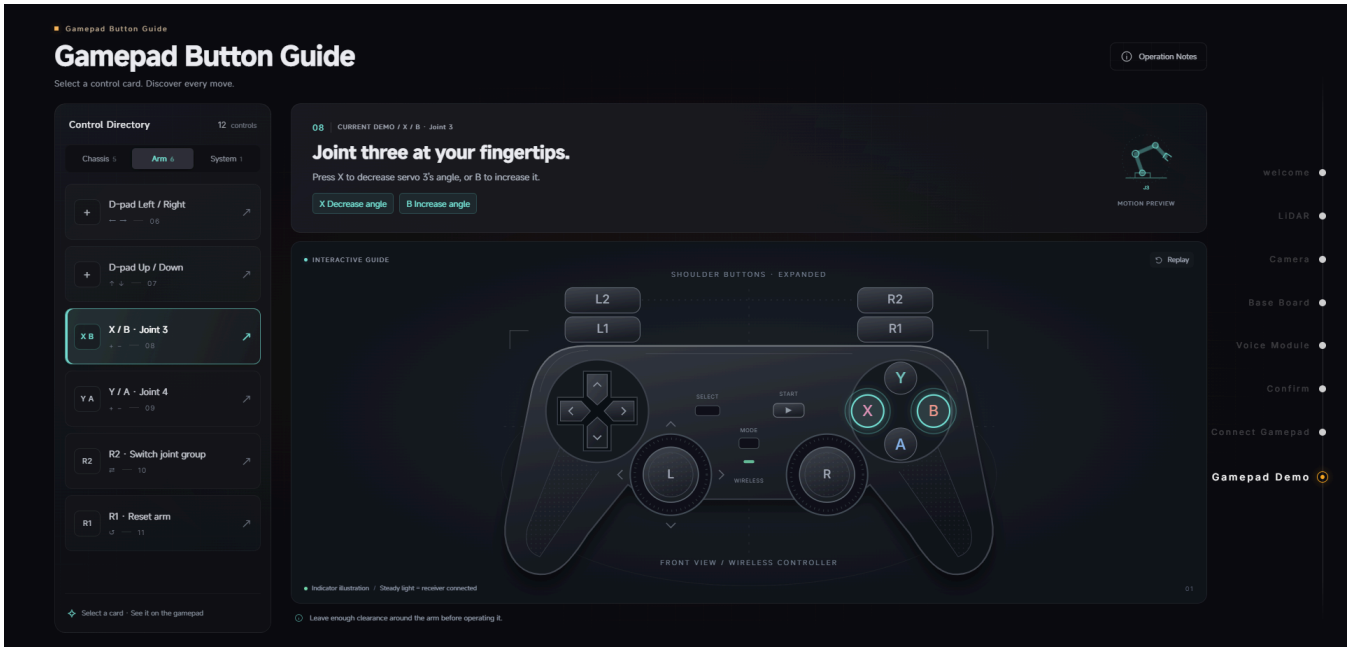
- Wait 1–2 minutes for the program to be ready, press the START button on the gamepad, and after hearing the "beep-beep-beep" prompt, gamepad control takes effect.



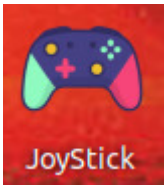
- Follow the gamepad key mapping instructions on the page to quickly experience gamepad control.



- You can quickly view the corresponding key functions by clicking the key cards on the left or the visualized gamepad keys.



- In subsequent use, you can also quickly launch the gamepad function via the launch icon on the VNC desktop. If it needs to work with other functions such as mapping and navigation, **note that you need to start the other programs first**, and then start the gamepad control program last.



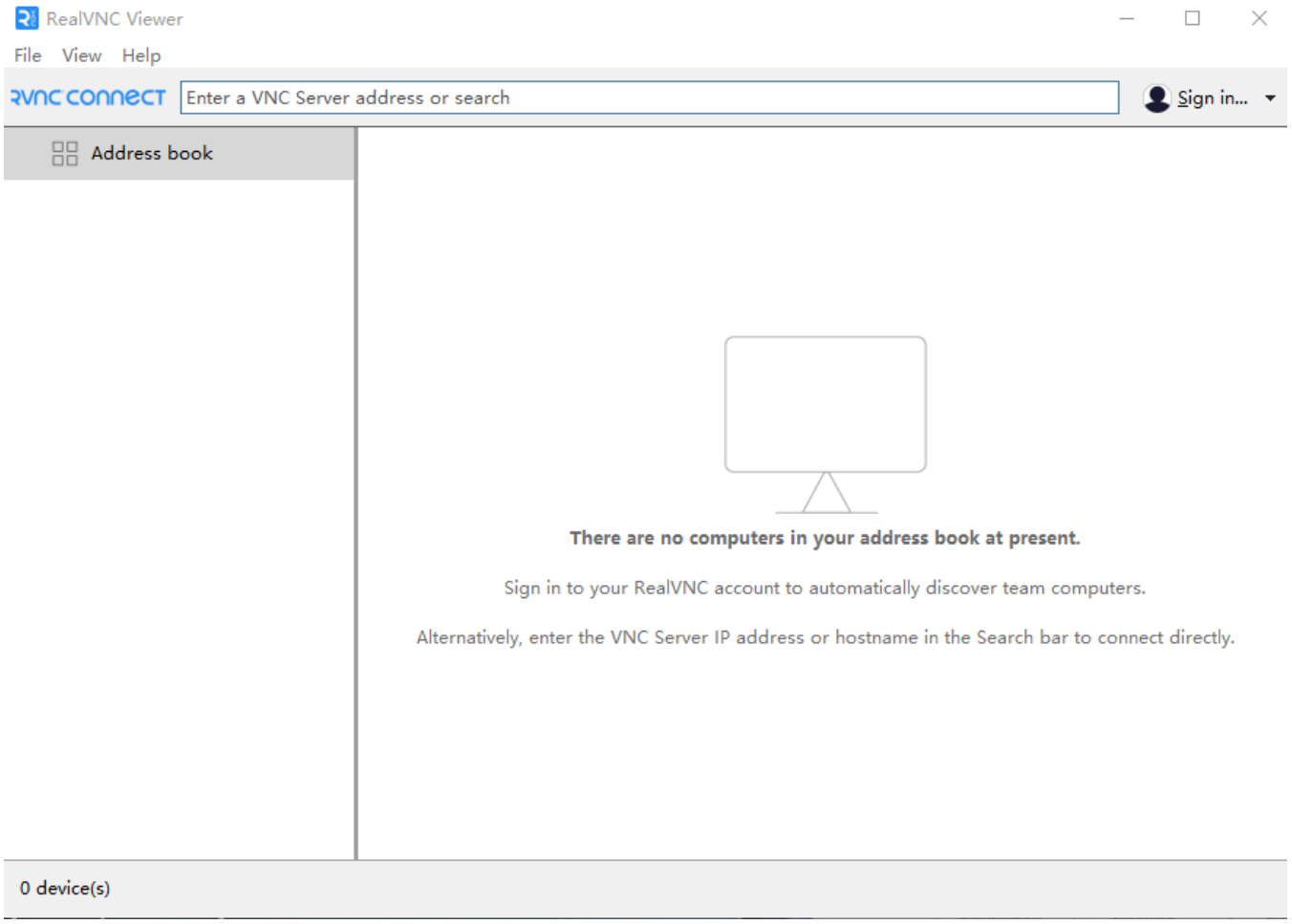
2. How to Connect to the Vehicle

2.1 VNC Connection

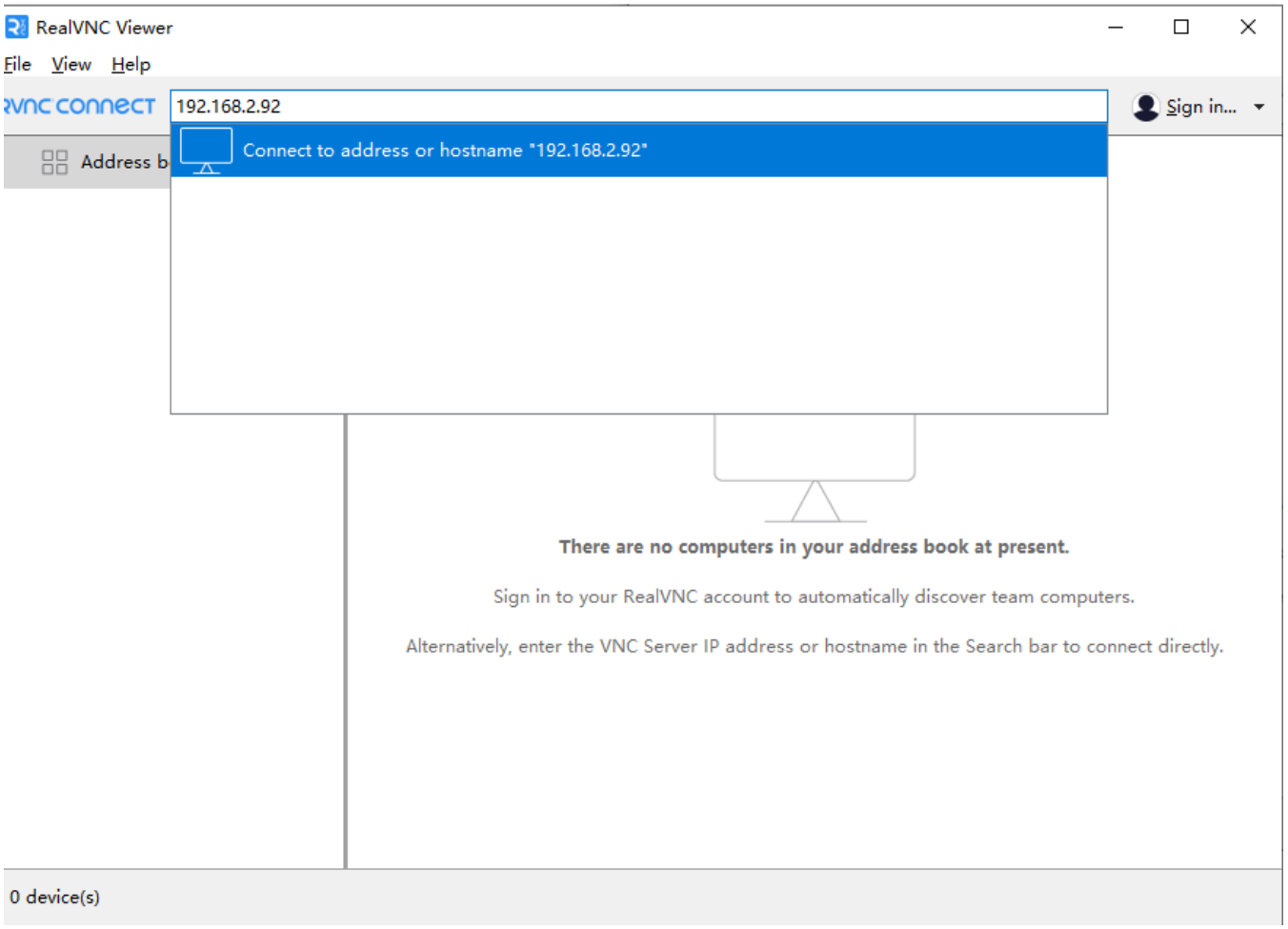
> **Tip**
>
> - There are many types of VNC connection tools. Here, a recommended tool for Windows is provided. Users of macOS and Linux can choose other VNC connection tools as needed; the tool is not unique.
> - For the first connection, you can connect your computer to the vehicle hotspot, and then log in and connect via the IP: **192.168.1.11**

- VNC allows users to remotely access and control the desktop environment of another computer over the network. Therefore, when we need to access the car's desktop environment, for example, to launch image display, we can use VNC to connect and log in to the car. The VNC download address is as follows: [Download VNC Viewer by RealVNC®]

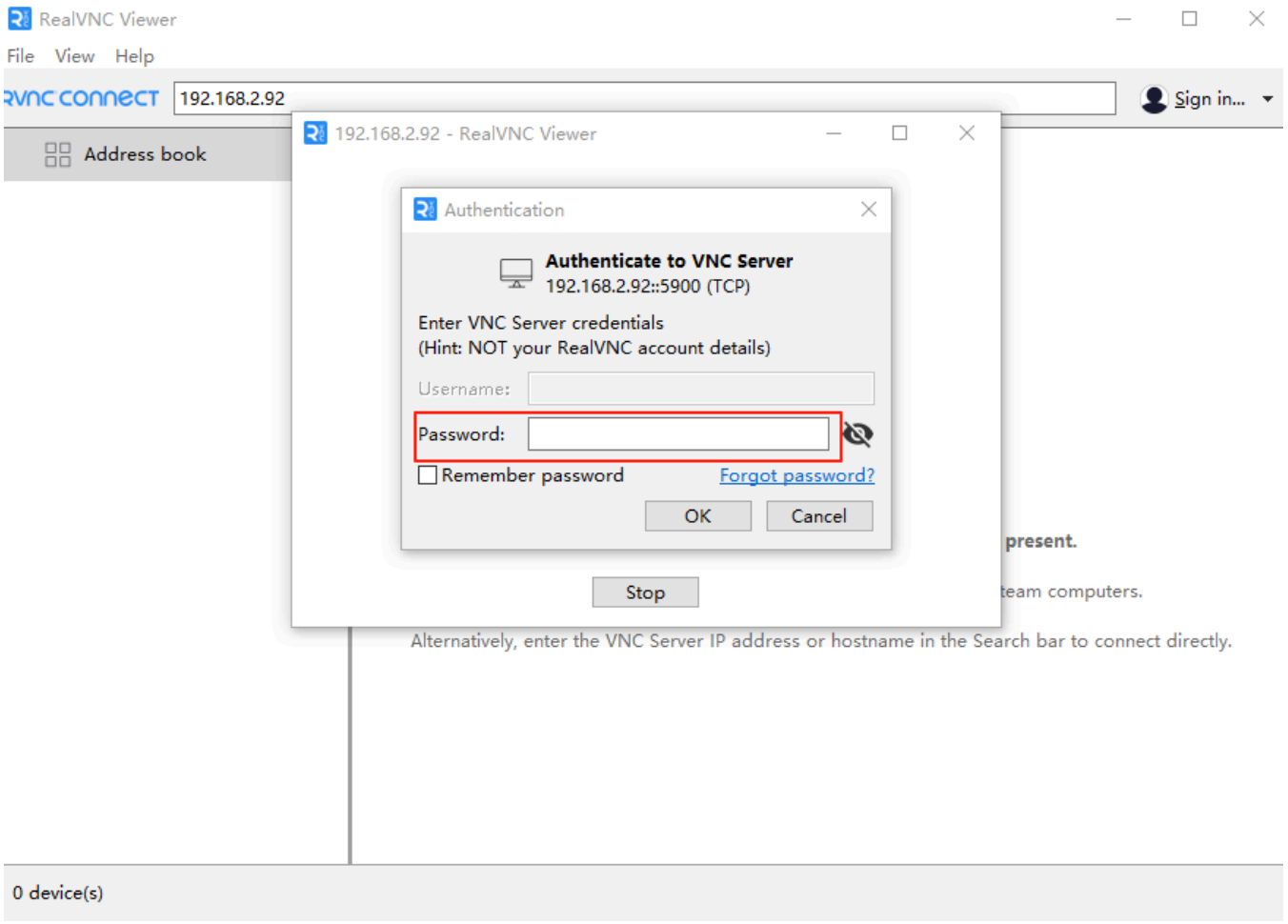
Download and install according to your computer version. After successful installation, double-click to open it. The screen displayed is as follows.



- Enter the IP address of the car. Here, my IP address is 192.168.2.92, as shown in the figure below.



- Then press Enter.



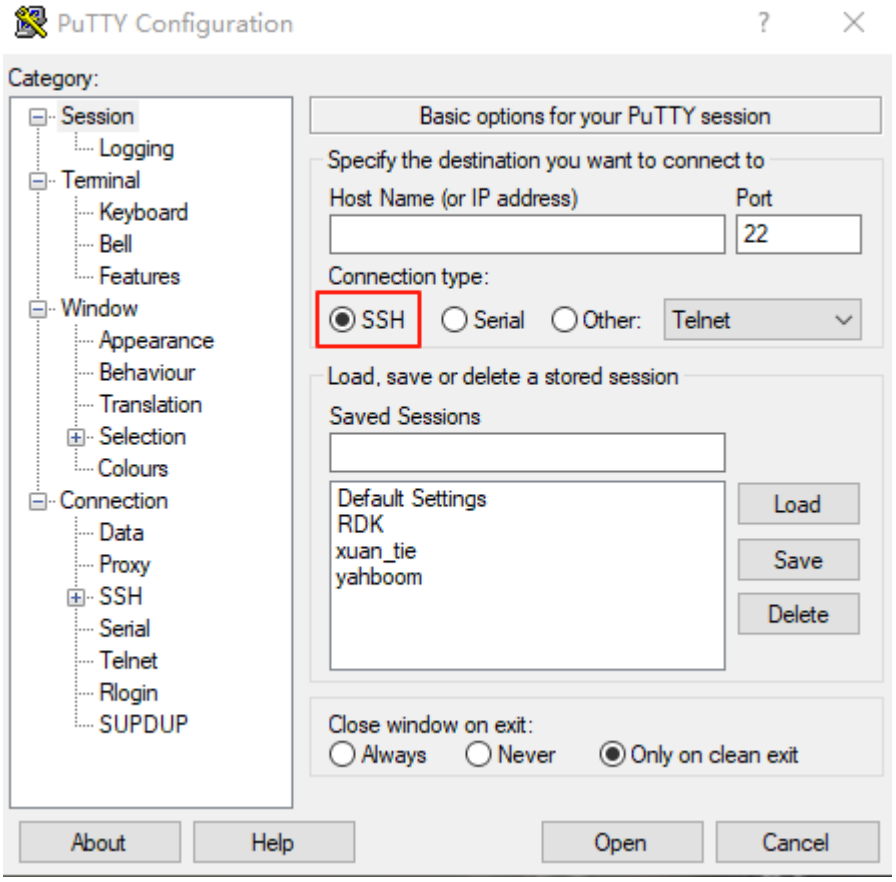
- Enter the username in Username and the password in Password. The password for all main boards is yahboom. Then click OK to enter the desktop, as shown in the figure below.



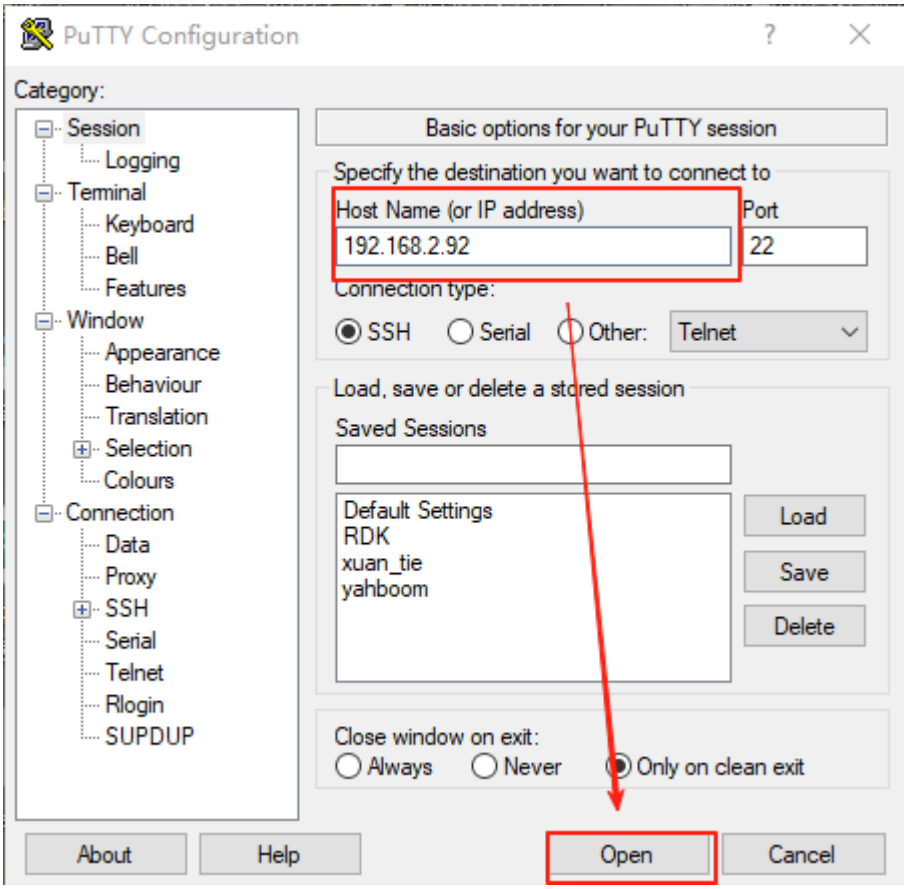
2.2 SSH Connection

> **Note**
>
> There are many types of SSH connection tools. Here, a recommended SSH connection tool for Windows is provided. Users of macOS and Linux can choose other SSH tools as needed.

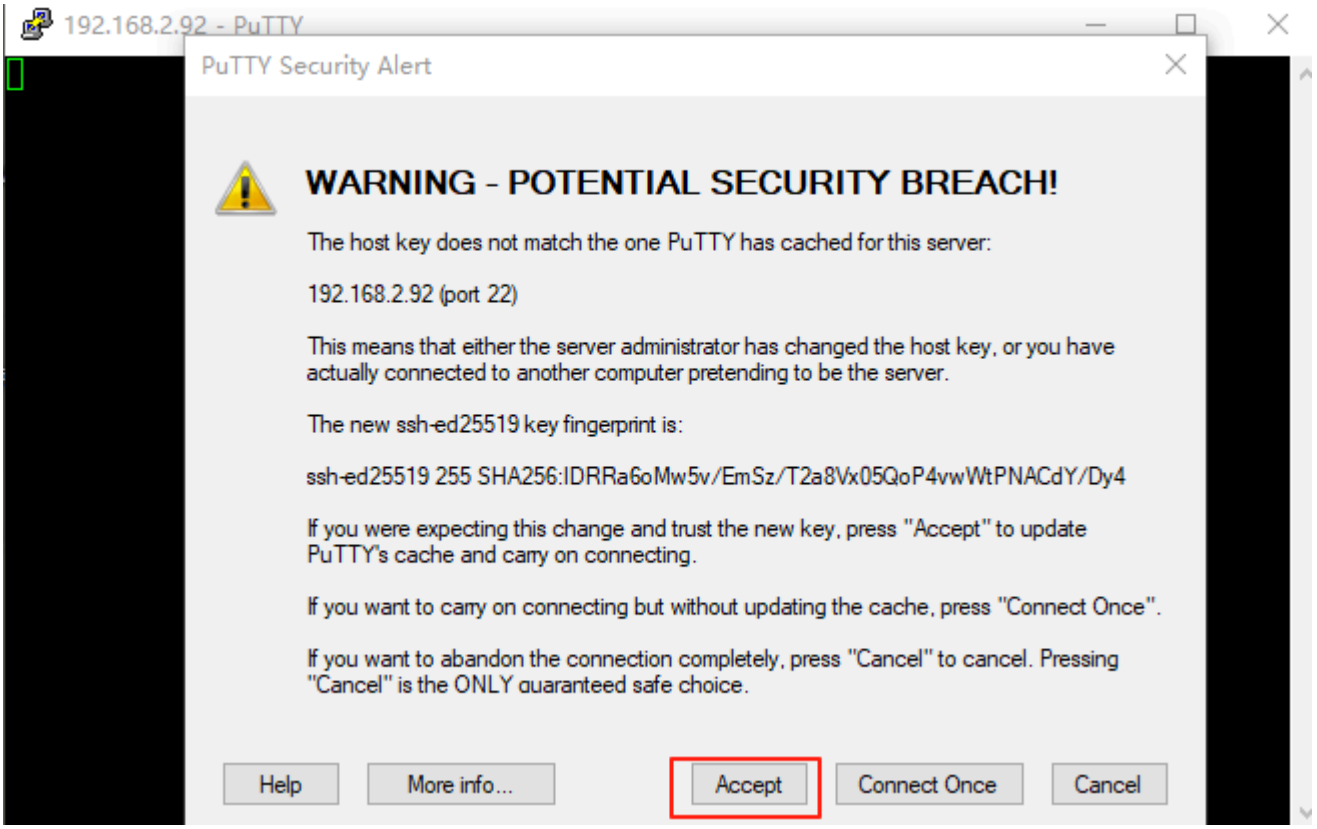
- Use PuTTY, MobaXterm, or other SSH login tools to connect to the car. Here, PuTTY is used as an example. The PuTTY download address is as follows: Download PuTTY: latest release (0.83)
- Select and install according to your computer version. After successful installation, double-click to open it. The PuTTY interface is shown in the figure below.



Here, select **SSH**, and then enter the IP address displayed on the OLED in the Host Name(or IP address) field. Here, my IP address is 192.168.2.92, so enter the IP address as shown in the figure below.



Then click Open. A terminal interface and a pop-up window will appear. Click Accept to accept, as shown in the figure below.

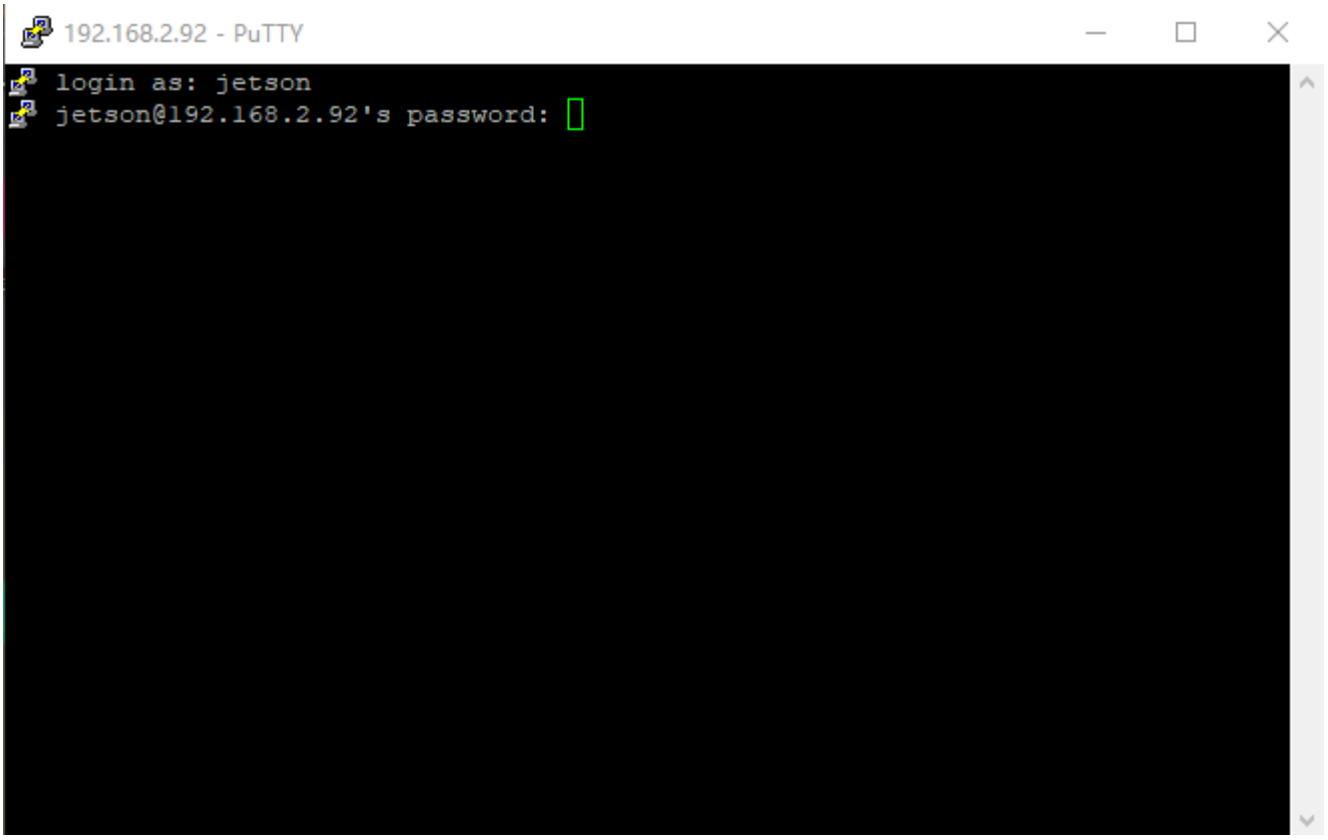


- Then the terminal will display **login as**. Enter the username of the car's main board here, and then press

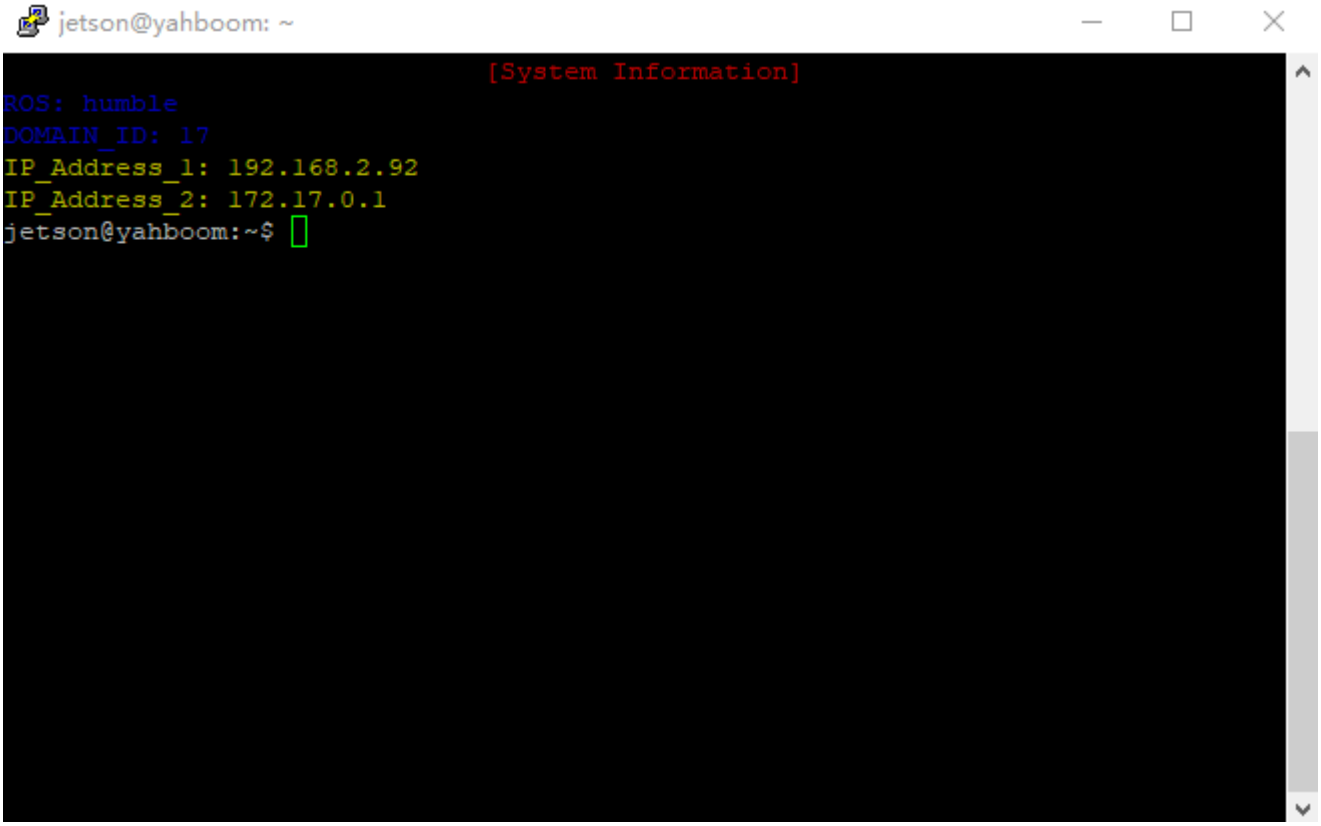
Enter. Next, you will be prompted to enter the password. Enter the password. The username and password for each main board are as follows:

Main Board	Username	Password
Raspberry Pi 5	pi	yahboom
RDK X5	sunrise	yahboom
Orin-Nano	jetson	yahboom
Orin-NX	jetson	yahboom

- Here, the Orin-Nano main board is used as an example for demonstration. Enter jetson, and then press Enter. Then enter the password. **Nothing is displayed while entering the password**. Enter yahboom, and then press Enter.



The interface after successfully connecting to the car is as follows.



- After that, you can operate the vehicle through the SSH terminal. SSH is suitable for login when a graphical interface is not required. Its smoothness is much higher than that of VNC, making it suitable for CLI command operations.

3. Connecting the Vehicle to Wi-Fi

> **Important**
>
> - Regardless of which tool is used to connect to the vehicle, **the computer and the car must be on the same local area network**.
> - The hotspot that starts automatically at the factory cannot connect to the Internet. During normal use, the vehicle connects to Wi-Fi, and the computer is used to operate within the same local area network.

3.1 How to Obtain the Robot Vehicle IP Address

1. **Check the OLED screen on the body directly (most intuitive)**:
    - After the car is powered on, the OLED screen on the body will display the IP address assigned to the current network interface (such as `192.168.12.57` assigned by the router or the built-in hotspot IP `192.168.1.11`).
2. **Connect to the car's built-in hotspot (offline, no router required)**:
    - On the computer, search for and connect to the car hotspot `ROSMASTER` in the Wi-Fi list (password `12345678`).
    - After a successful connection, the car's static IP is fixed at: `192.168.1.11`.
3. **Wired Ethernet connection (most recommended, most stable connection)**:

- Insert one end of the Ethernet cable into the car's main board network port and connect the other end to the router. The car will automatically obtain a LAN IP via DHCP and display it on the OLED screen in real time.

3.2 Connecting the Robot Vehicle to an External Wi-Fi

The car's built-in hotspot has no external network access capability. If you need to use AI large models, install software packages online, or update the system later, it is recommended to connect the car to the home router's Wi-Fi.

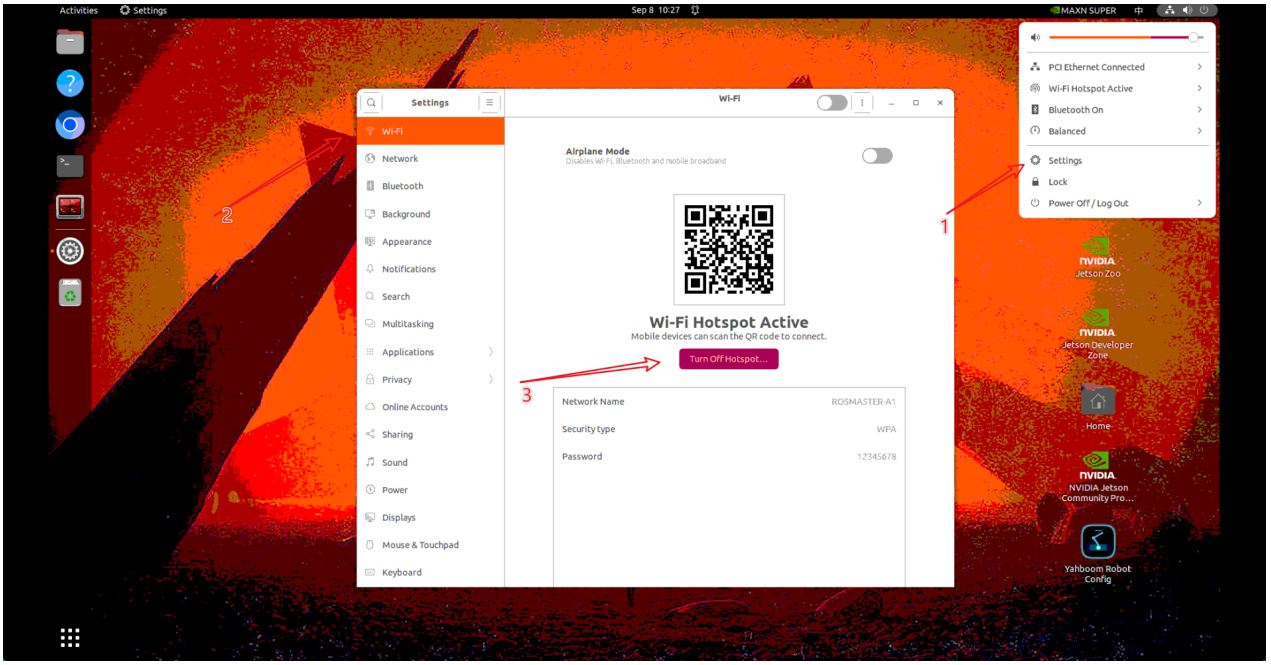
There are two common ways to connect to an external Wi-Fi:

Method 1: Connect via the VNC graphical interface with an Ethernet cable (Recommended)

1. Insert a wired Ethernet cable into the car's main board, read the IP address from the OLED screen on the body (for example, `192.168.12.57`), and use VNC Viewer to log in to the car via remote desktop (for detailed VNC login methods, see Section 2.1). After a successful login, the desktop is as follows:



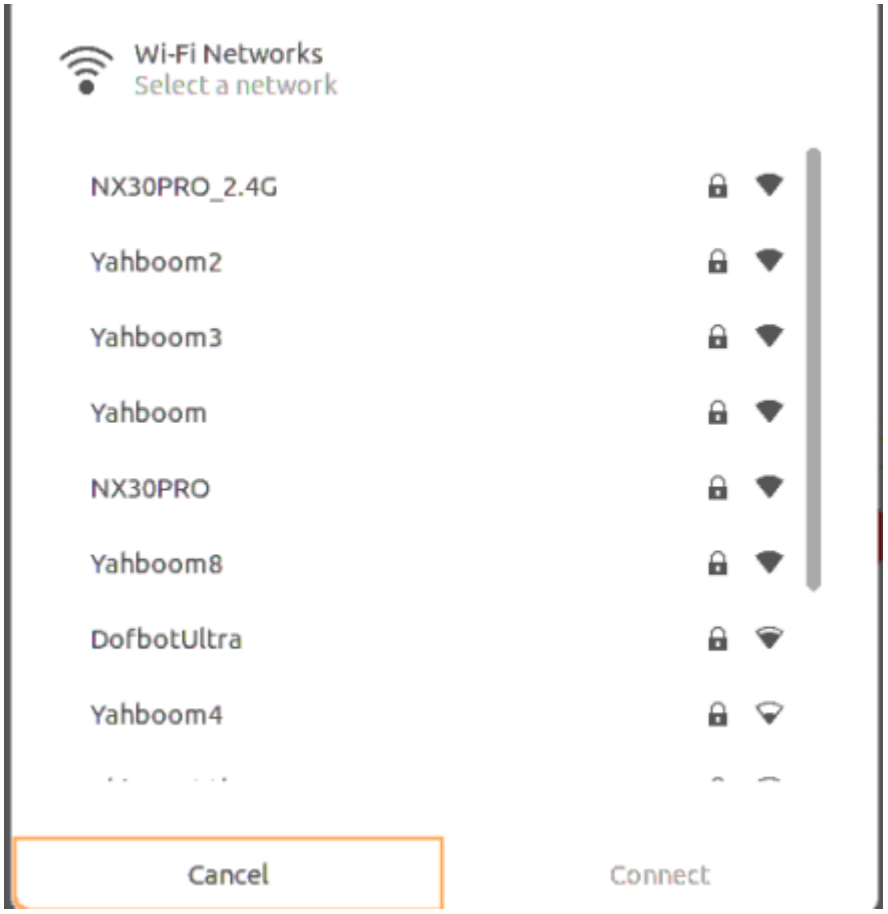
2. Open System Settings (Settings) and enter the Wi-Fi settings interface. **First turn off the car's built-in hotspot** so that the network adapter can scan for external Wi-Fi normally:

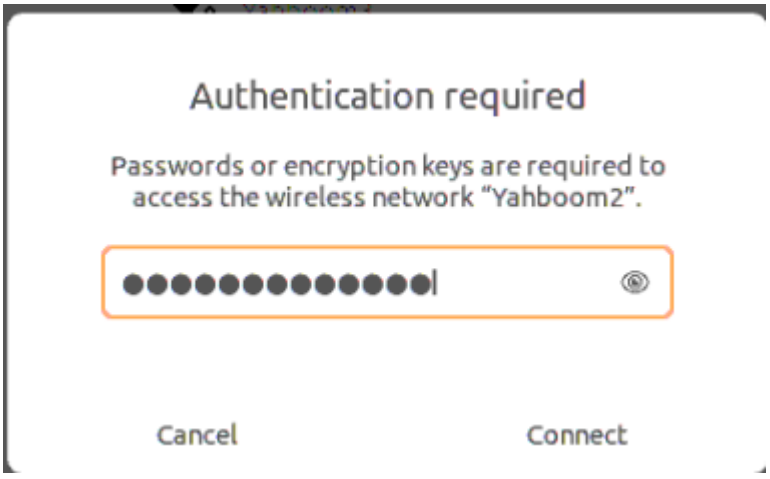


3. Click the network icon in the upper right corner of the desktop, and click **Select Network** in the drop-down menu:

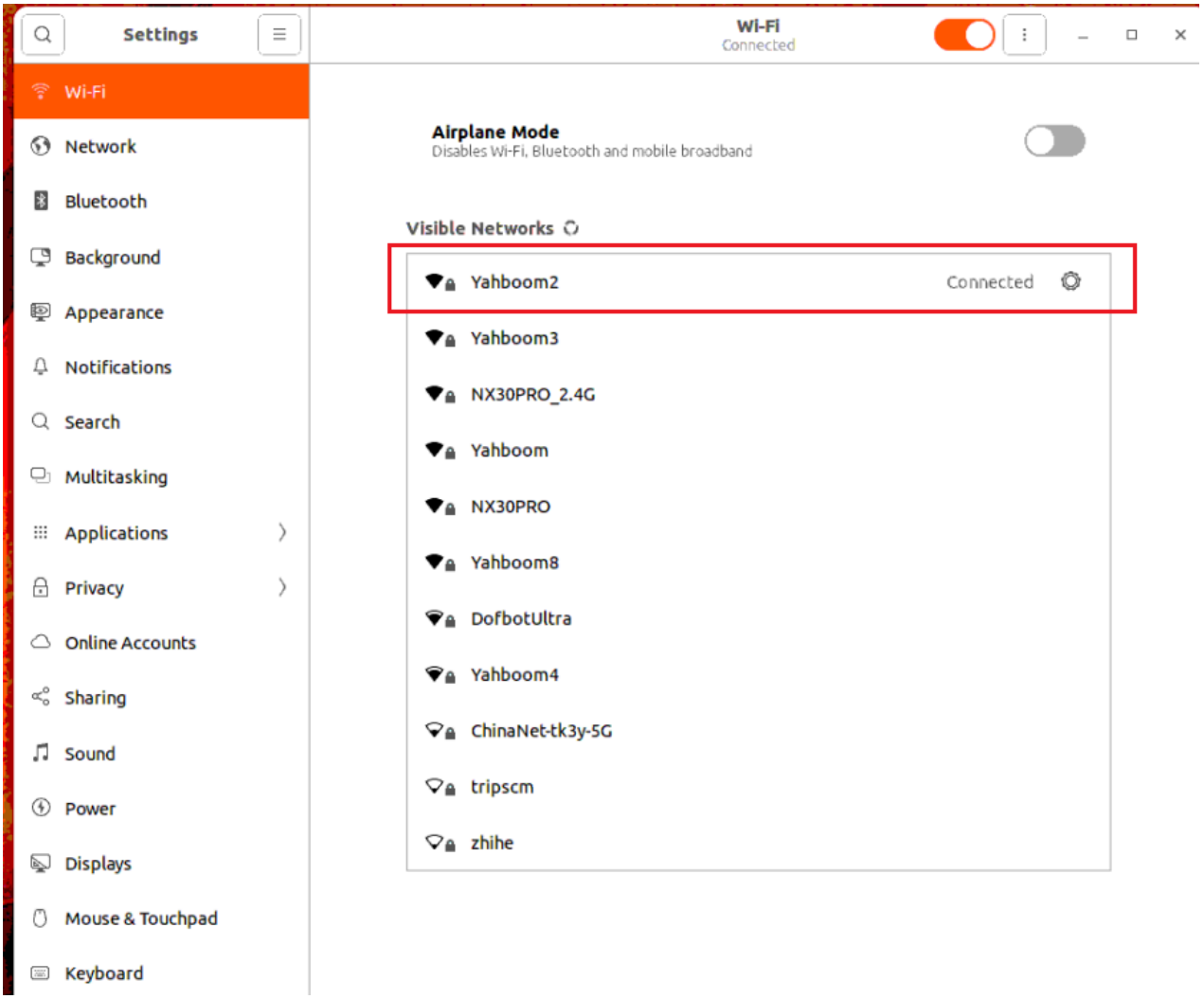


4. Select the name of the external Wi-Fi you want to connect to, enter the password, and click Connect:





5. After a successful connection, the network icon will show the connected state. At this time, unplug the Ethernet cable, and the OLED screen will also update to the new Wi-Fi IP address. You can then use this new IP for remote connections.



Method 2: Without an Ethernet cable, configure Wi-Fi via the hotspot IP and modify the Wi-Fi priority for automatic connection

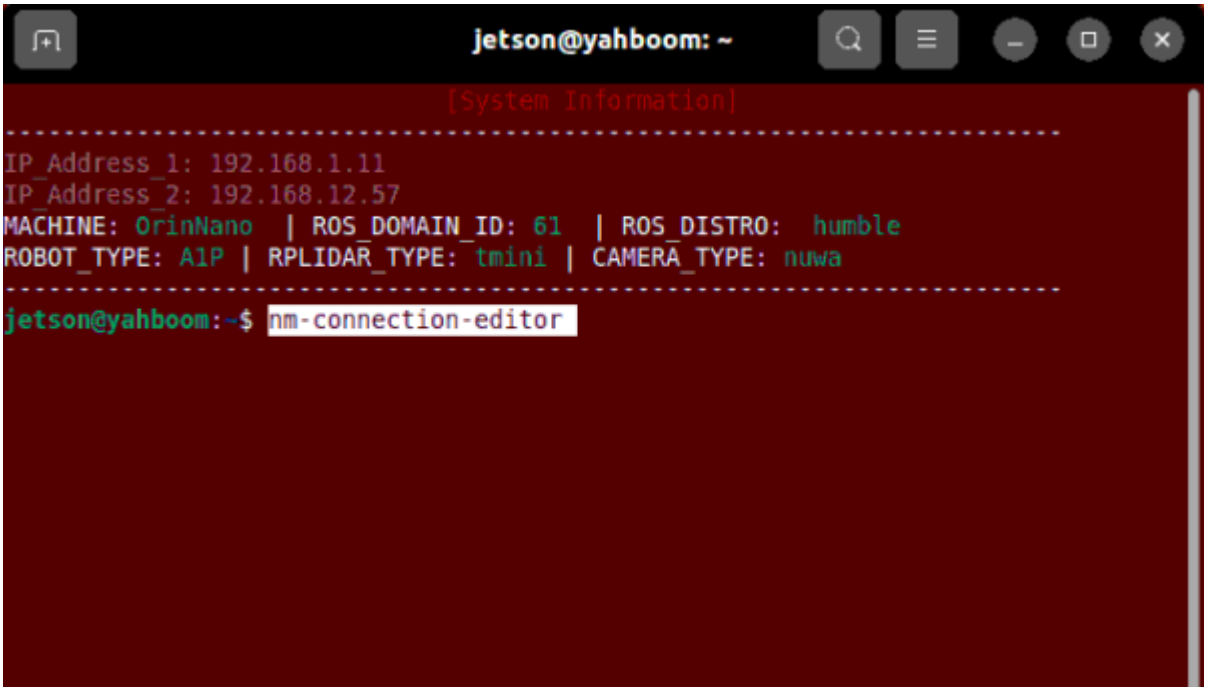
If there is no router Ethernet cable, you can connect your computer directly to the car's default Wi-Fi hotspot:

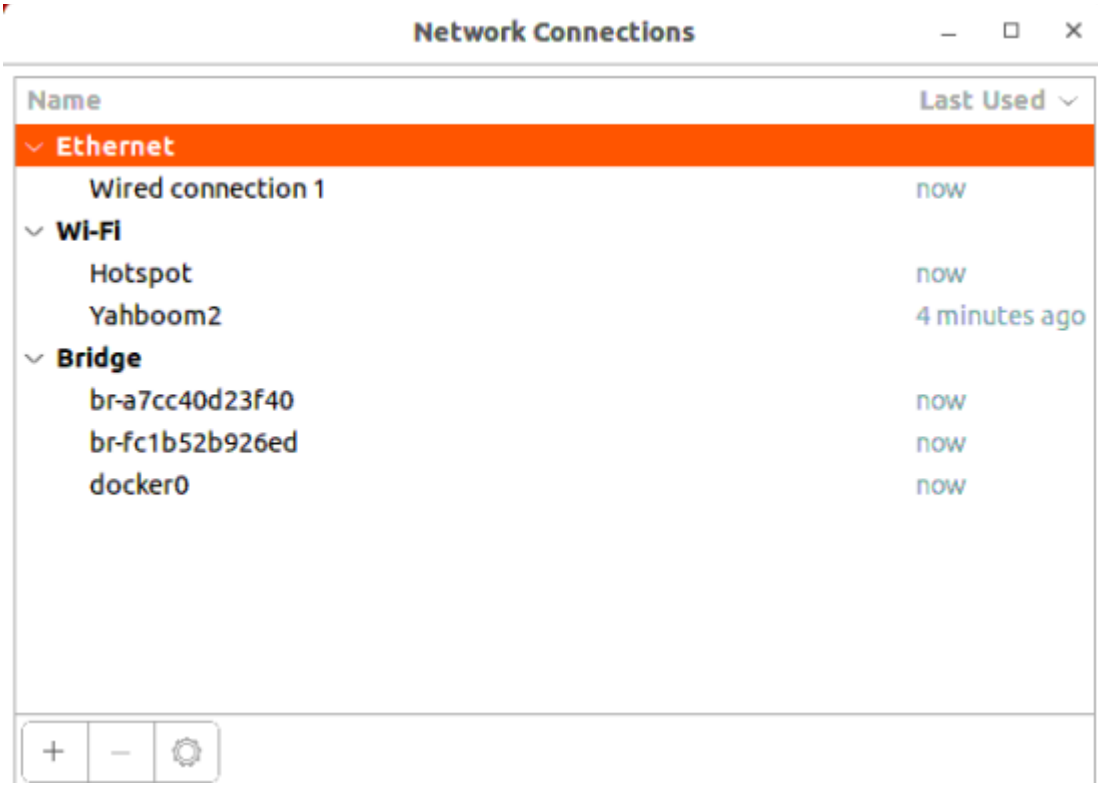
1. Connect the computer to the car hotspot `ROSMASTER` (password `12345678`), and use VNC Viewer to connect to the default hotspot IP: `192.168.1.11`.



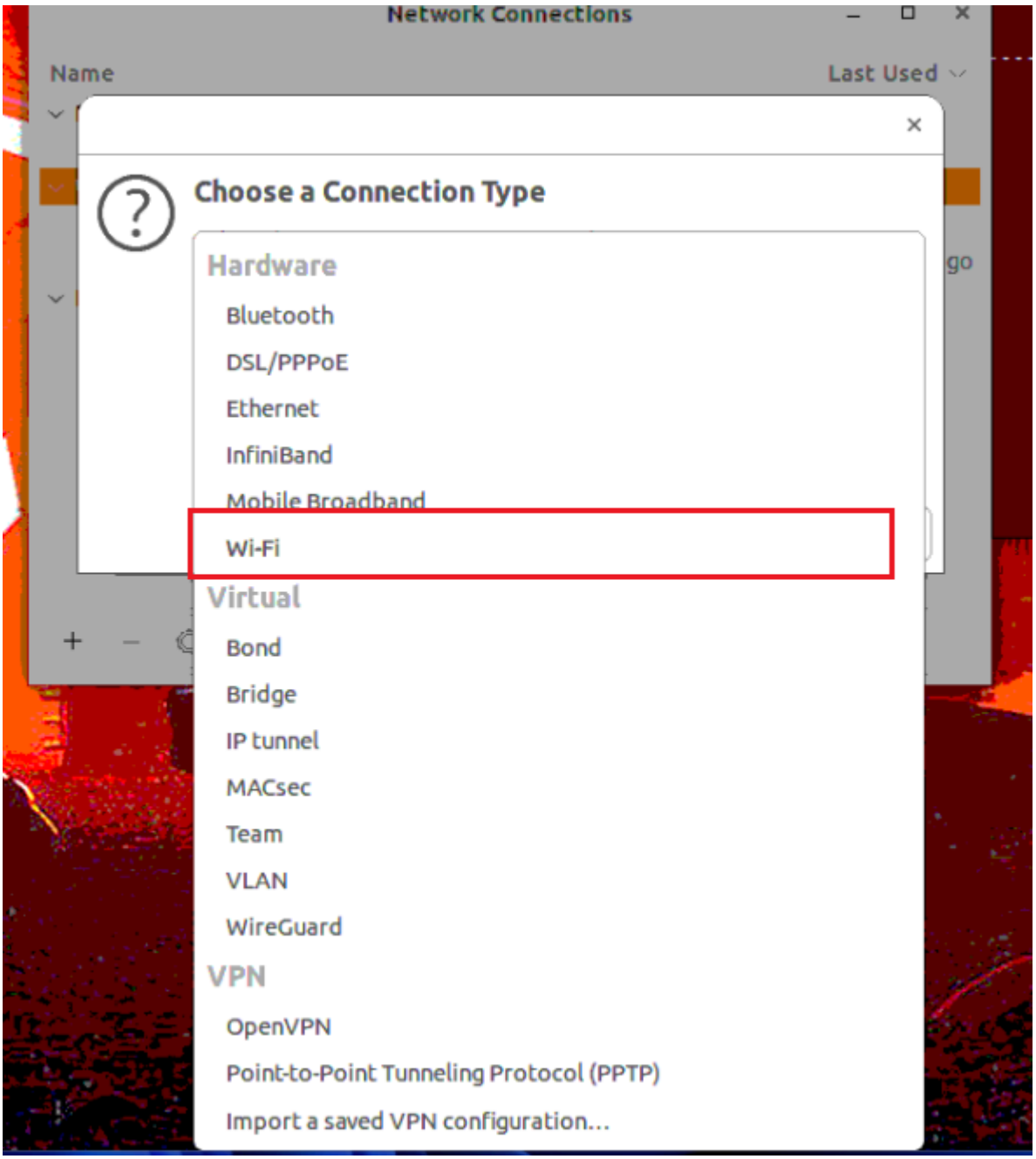
2. Open a terminal on the desktop and enter the network configuration tool command:

```
nm-connection-editor
```

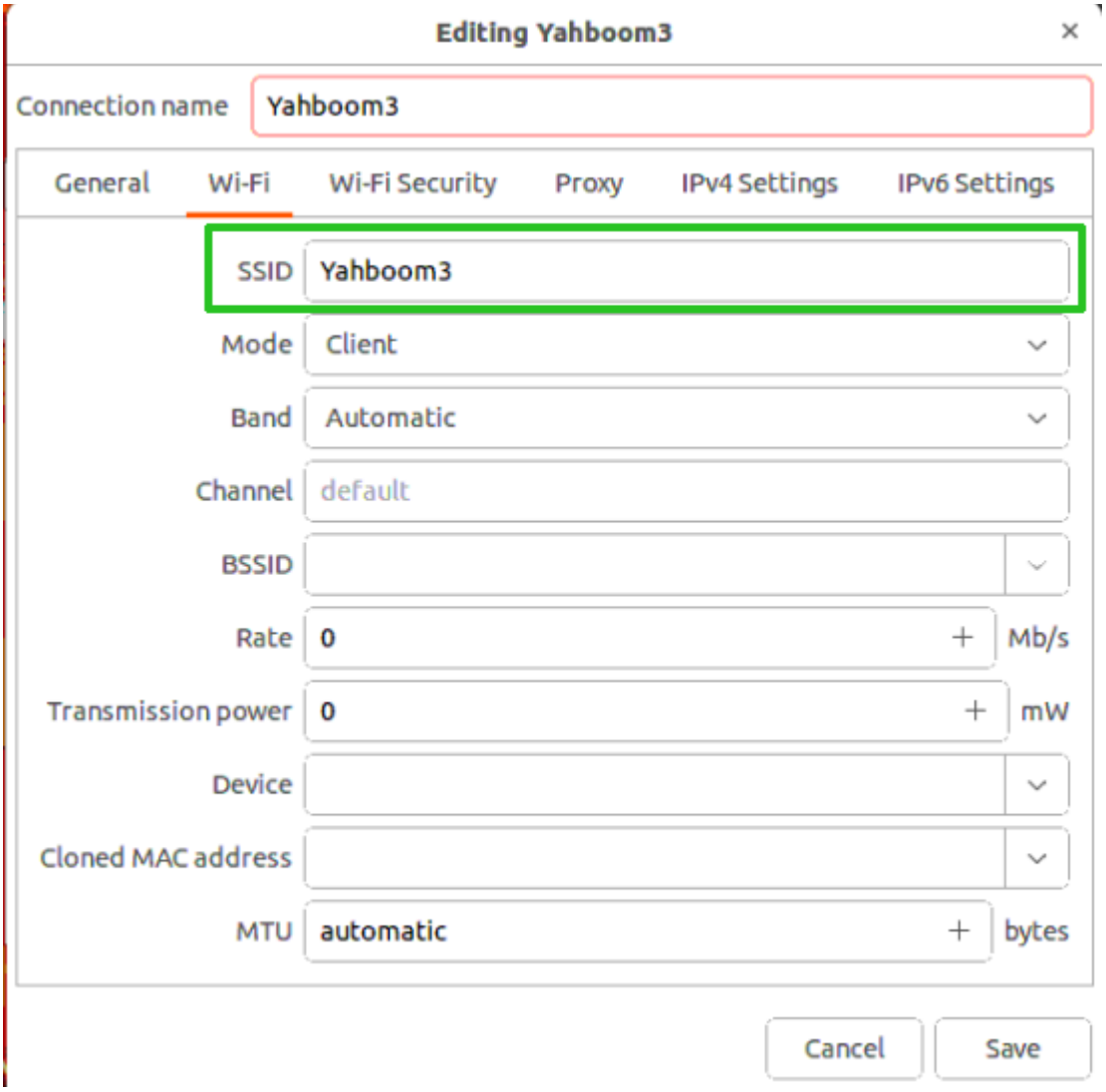




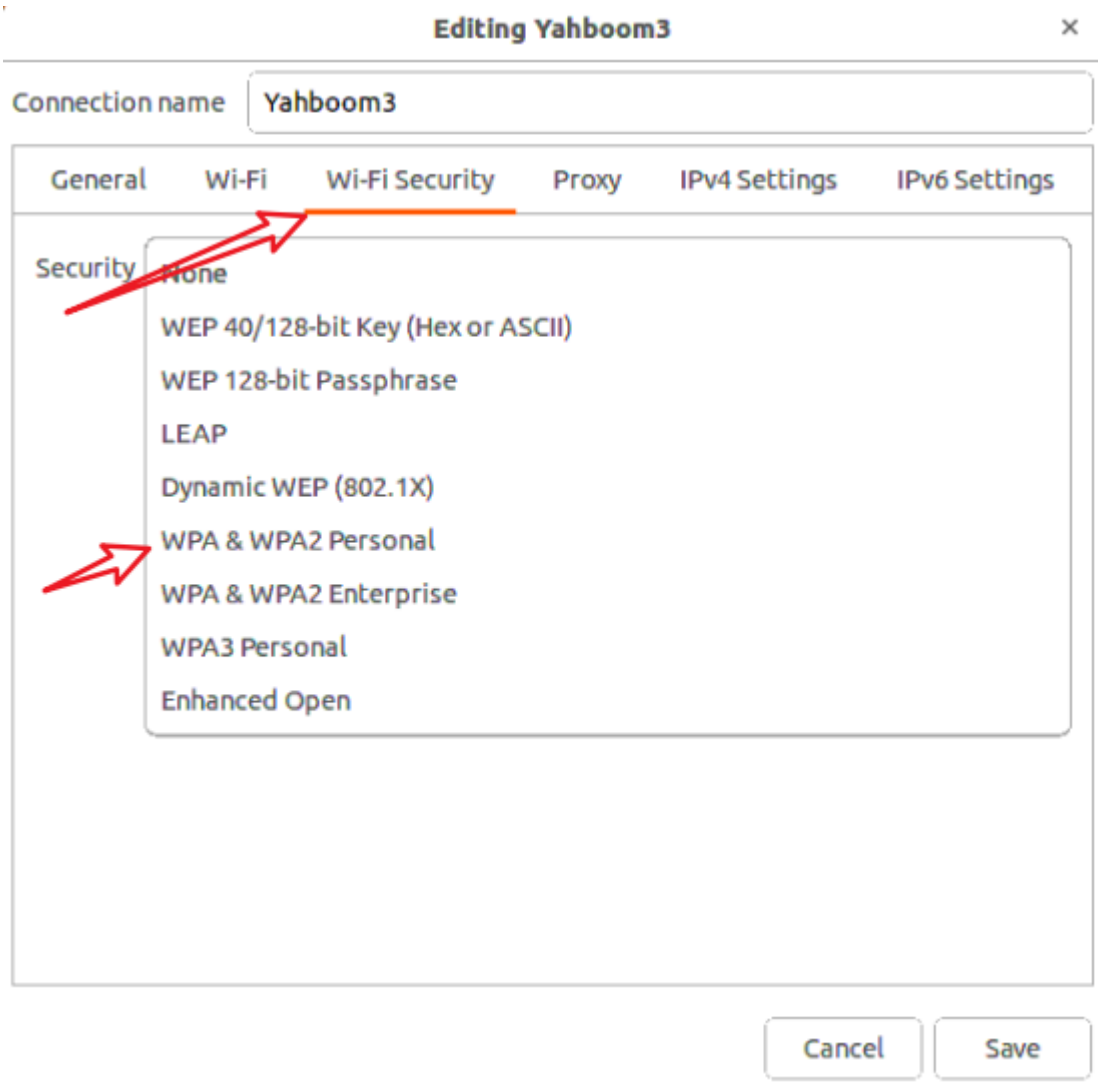
3. In the Network Connections window that appears, click 【+】 in the lower left corner to create a new connection, select **Wi-Fi** as the connection type, and click **Create**:

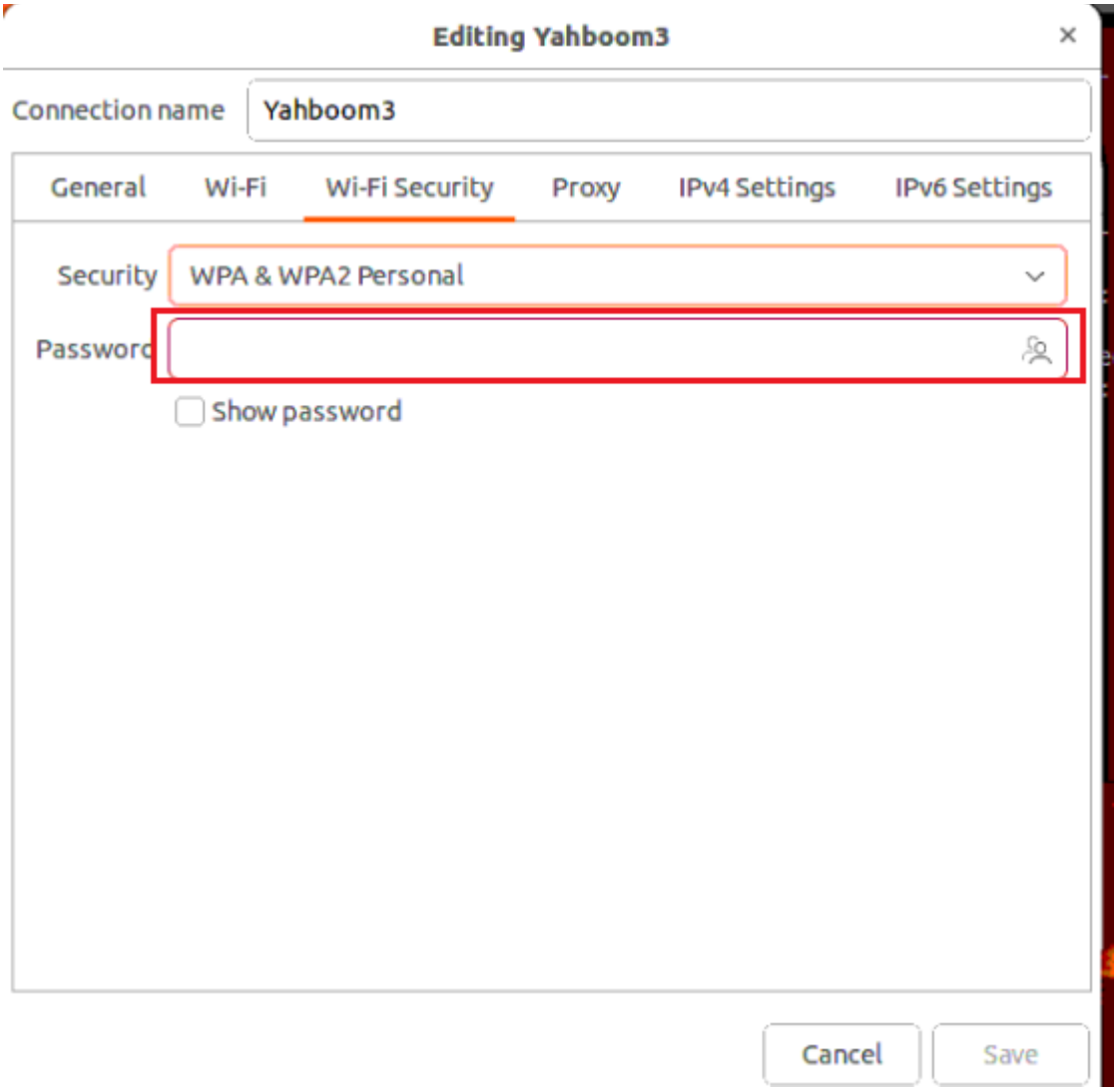


4. In the **SSID** field of the **Wi-Fi** tab, enter the name of the router Wi-Fi you want to connect to:

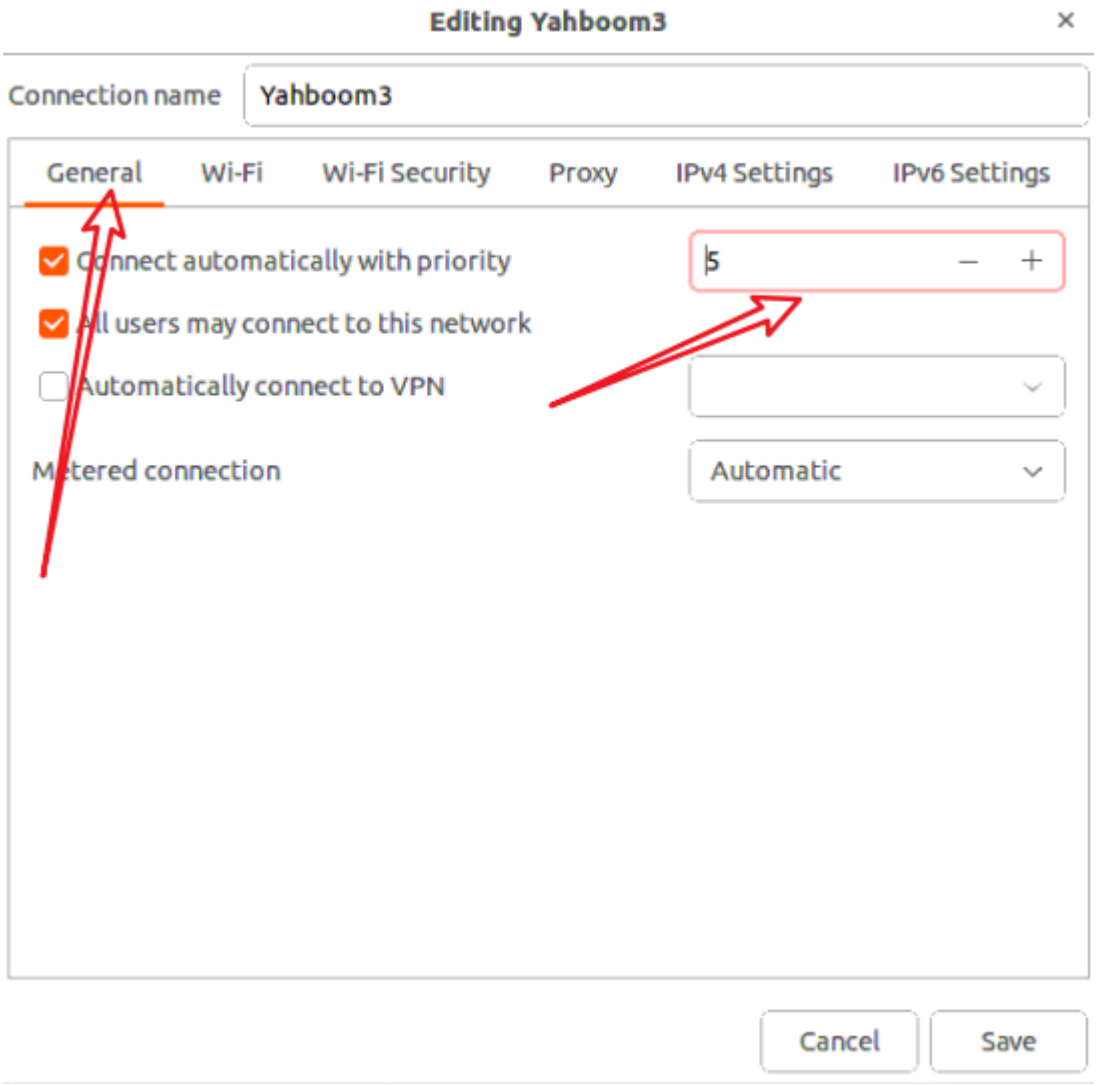


5. Switch to the **Wi-Fi Security** tab, select **WPA & WPA2 Personal** as the security type, and enter the connection password for the Wi-Fi in the **Password** field:

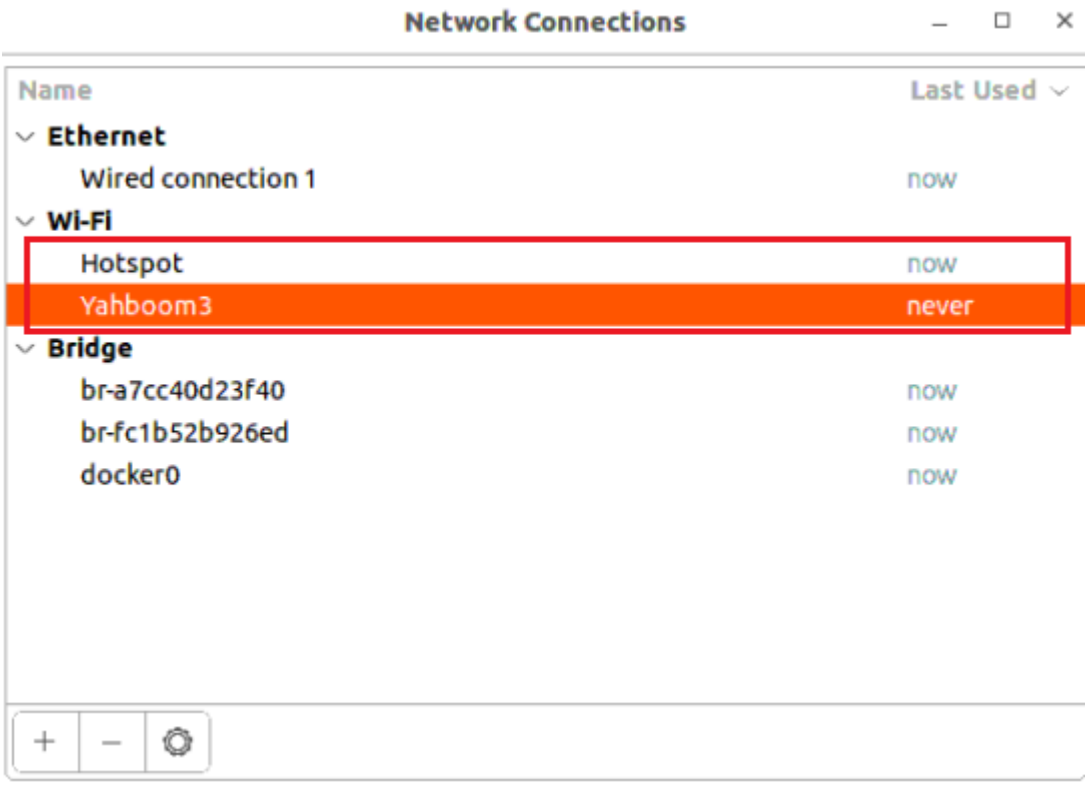

Editing Yahboom3
Connection name
Yahboom3
General
Wi-Fi
Wi-Fi Security
Proxy
IPv4 Settings
IPv6 Settings
Security
None
WEP 40/128-bit Key (Hex or ASCII)
WEP 128-bit Passphrase
LEAP
Dynamic WEP (802.1X)
WPA & WPA2 Personal
WPA & WPA2 Enterprise
WPA3 Personal
Enhanced Open
Cancel
Save



6. Switch to the **General** tab, check "Connect automatically", and increase the **Priority (boot connection priority)** value (for example, set it to `10`; the larger the value, the higher the priority):



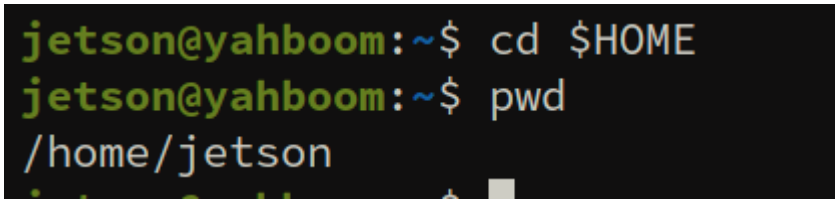
7. Click **Save** in the lower right corner to save. Similarly, you can double-click the hotspot connection in the list to view the hotspot's priority. It is sufficient as long as the external Wi-Fi priority is greater than the hotspot priority:



8. After configuration is complete, close the window and restart the main board. After the car is powered on, it will automatically prioritize connecting to the configured external Wi-Fi.

4. Tutorial Symbol Description

- The ~ and $HOME symbols appearing in the tutorial represent the user directory of the main board in **Linux**, for example:



- The < xxx > symbol is a placeholder, representing custom content.

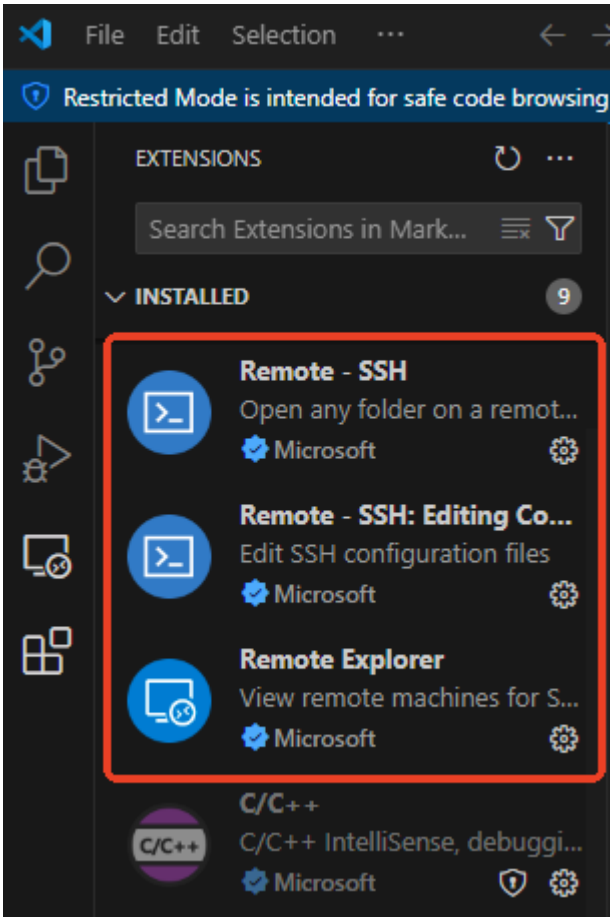
5. Viewing Source Code and Building Packages

> **Important**
>
> - Visual Studio Code here is a recommended tool. If you have your own preferred IDE, you can use any tool.
> - Visual Studio Code is a lightweight, efficient, extensible IDE that integrates core development tools. Official website: https://code.visualstudio.com/

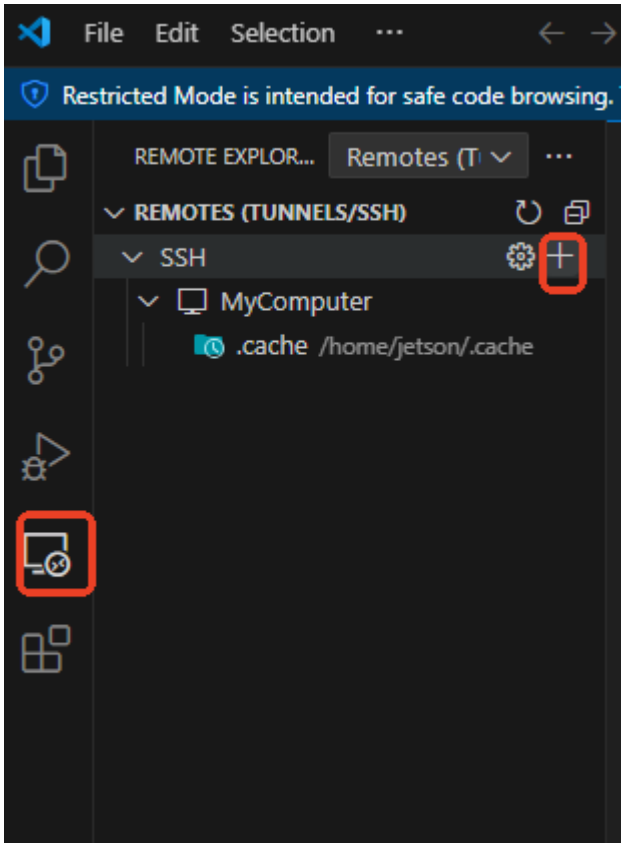
5.1 Viewing Code

- If Visual Studio Code is not installed, you can download and install it via the link above. Launch Visual

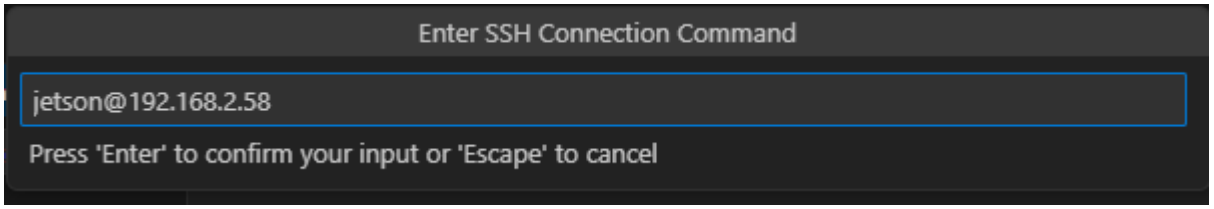
Studio Code, click Extensions on the left, and install the three remote connection extensions: Remote-SSH, Remote - SSH: Editing Configuration Files, and Remote Explorer.



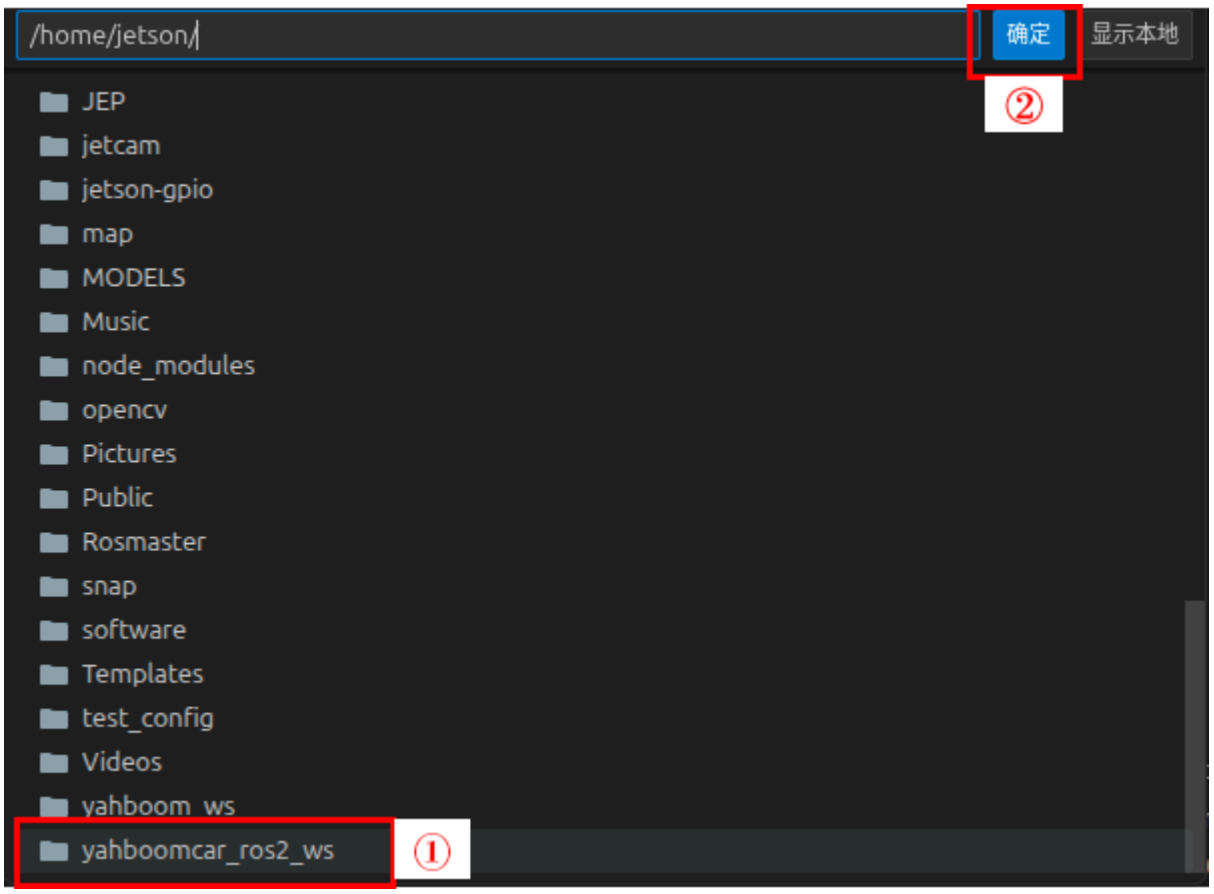
- Then click Remote Connection on the left—click New Remote.



- In the input box that appears, enter username@IP address according to the main board. After pressing Enter, enter the user password yahboom to log in to the vehicle.

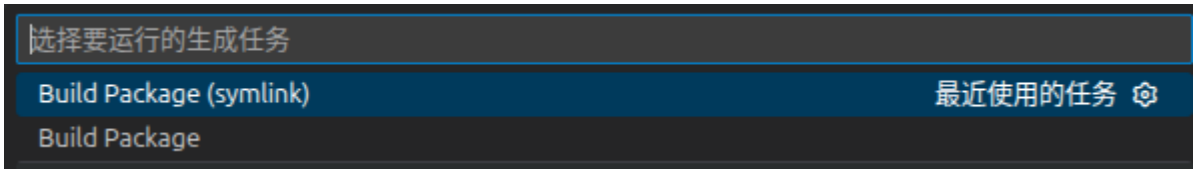


- Then click Open Folder. In the window that appears, you can select the folder you need to view. Here, yahboomcar_ros2_ws is used as an example, and then click OK.



5.2 Building Packages

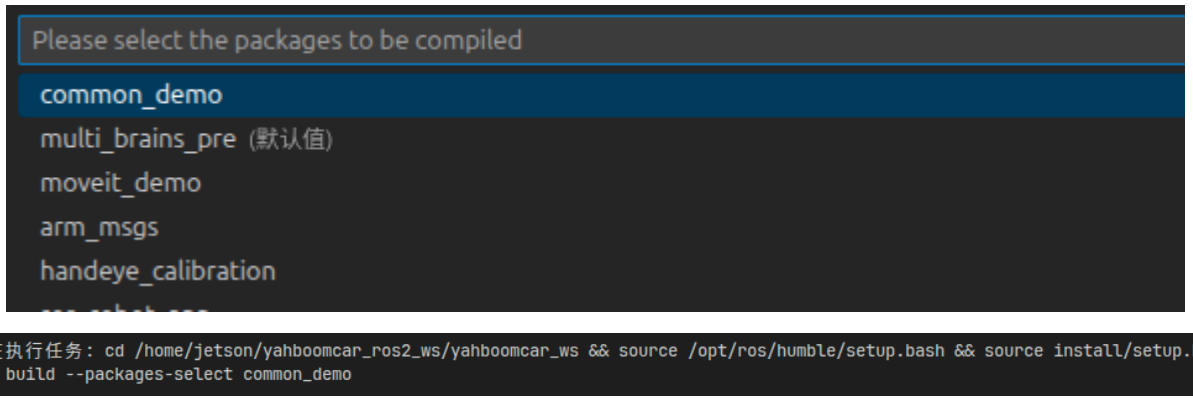
- In the VS Code interface, press the shortcut Ctrl+Shift+B to bring up the build options window.



- Two build methods are provided: link build and normal build.
    - Link build: yahboomcar_bringup, yahboomcar_description, m1pro_moveit_config (after the configuration files and launch files of the packages are modified, the changes take effect directly without rebuilding).

    - Normal build: the source code of other packages needs to be built before taking effect.

- After selecting a build option, select the function name to be built, and then press Enter to confirm. The command will be executed automatically to build the corresponding package.




```
* 正在执行任务: cd /home/jetson/yahboomcar_ros2_ws/yahboomcar_ws && source /opt/ros/humble/setup.bash && source install/setup.bash &&
colcon build --packages-select common_demo
```

> **Important**
>
> All configuration files and launch files in the yahboomcar_bringup function take effect immediately after being modified and saved, without rebuilding. Building is only used when the user needs to modify the source code for development requirements.